

# Proteus Wiring Specifications

## 200 canonical circuits with complete component-to-pin connectivity

- This document is an implementation specification, not a theory guide.
- Each circuit uses generic Proteus primitives matching the earlier component-count workbook.
- For every placed component, every electrical pin is assigned to exactly one named net.
- To wire a circuit, place the listed components and join every pin appearing on the same net.
- Ground terminals, probes, oscilloscopes, meters, graph instruments, connectors and net labels are not counted as components.
- OPAMP is treated as one generic five-pin op-amp section: IN+, IN-, OUT, V+ and V-.
- LM317T physical pins are ADJ(1), OUT(2) and IN(3). POT-HG pins are 1, W and 3.
- A POT-HG used as a rheostat intentionally ties its wiper to one end; both pins are still explicitly accounted for.
- Source + and - pins are fully listed. Negative-rail VDC sources are oriented with + at GND and - at V-.

### Pin conventions

| Proteus part | Pins used in this document |
|--------------|----------------------------|
| RES          | 1, 2                       |
| CAP          | 1, 2                       |
| CAP-ELEC     | +, -                       |
| INDUCTOR     | 1, 2                       |
| POT-HG       | 1, W, 3                    |
| DIODE        | A, K                       |
| ZENER        | A, K                       |
| LED          | A, K                       |
| NMOSFET      | G, D, S                    |
| OPAMP        | IN+, IN-, OUT, V+, V-      |
| LM317T       | ADJ(1), OUT(2), IN(3)      |
| VDC          | +, -                       |
| VSINE        | +, -                       |
| VPULSE       | +, -                       |
| IDC          | +, -                       |

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# Circuit 1: DC Series Resistor Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=N1      |
| R2  | RES          | 2.2 kOhm        | 1=N1; 2=GND      |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, R2.2                |
| VIN | V1.+, R1.1                |
| N1  | R1.2, R2.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 2: DC Parallel Resistor Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=GND     |
| R2  | RES          | 2.2 kOhm        | 1=VIN; 2=GND     |
| R3  | RES          | 4.7 kOhm        | 1=VIN; 2=GND     |

### Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, R1.2, R2.2, R3.2    |
| VIN | V1.+, R1.1, R2.1, R3.1    |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

### Circuit 3: Mixed Series-Parallel Resistor Network

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=N1      |
| R2  | RES          | 2.2 kOhm        | 1=N1; 2=N2       |
| R3  | RES          | 3.3 kOhm        | 1=N1; 2=N2       |
| R4  | RES          | 1 kOhm          | 1=N2; 2=GND      |

Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, R4.2                |
| VIN | V1.+, R1.1                |
| N1  | R1.2, R2.1, R3.1          |
| N2  | R2.2, R3.2, R4.1          |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 4: Resistor Voltage Divider

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm         | 1=VIN; 2=VOUT    |
| R2  | RES          | 10 kOhm         | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R2.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, R2.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 5: Loaded Voltage Divider

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm         | 1=VIN; 2=VOUT    |
| R2  | RES          | 10 kOhm         | 1=VOUT; 2=GND    |
| RL  | RES          | 10 kOhm         | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R2.2, RL.2          |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, R2.1, RL.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 6: Potentiometer Voltage Divider

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

POT-HG: 1 | Voltage source: 1 | Total: 2

| Ref | Proteus part | Value / setting | Every pin -> net     |
|-----|--------------|-----------------|----------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND         |
| RV1 | POT-HG       | 10 kOhm         | 1=VIN; W=VOUT; 3=GND |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RV1.3               |
| VIN  | V1.+, RV1.1               |
| VOUT | RV1.W                     |

PIN AUDIT: PASS - 5 of 5 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 7: Potentiometer Rheostat Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | POT-HG: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net  |
|-----|--------------|-----------------|-------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND      |
| RV1 | POT-HG       | 10 kOhm         | 1=VIN; W=N1; 3=N1 |
| RL  | RES          | 1 kOhm          | 1=N1; 2=GND       |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, RL.2                |
| VIN | V1.+, RV1.1               |
| N1  | RV1.W, RV1.3, RL.1        |

PIN AUDIT: PASS - 7 of 7 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 8: Wheatstone Bridge

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 10 V            | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=NL      |
| R2  | RES          | 1 kOhm          | 1=NL; 2=GND      |
| R3  | RES          | 1 kOhm          | 1=VIN; 2=NR      |
| R4  | RES          | 1.2 kOhm        | 1=NR; 2=GND      |

### Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, R2.2, R4.2          |
| VIN | V1.+, R1.1, R3.1          |
| NL  | R1.2, R2.1                |
| NR  | R3.2, R4.1                |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 9: Resistive Attenuator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 9 kOhm          | 1=VIN; 2=VOUT    |
| R2  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R2.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, R2.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 10: Resistor Current-Limiter Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| RL  | RES          | 100 Ohm         | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, RL.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 11: RC Charging Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting    | Every pin -> net |
|-----|--------------|--------------------|------------------|
| V1  | VPULSE       | 0-10 V, 1 s period | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm            | 1=VIN; 2=VOUT    |
| C1  | CAP          | 100 uF             | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, C1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 12: RC Discharging Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting    | Every pin -> net |
|-----|--------------|--------------------|------------------|
| V1  | VPULSE       | 10-0 V, 1 s period | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm            | 1=VIN; 2=VOUT    |
| C1  | CAP          | 100 uF             | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, C1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 13: RC Delay-On Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | LED: 1 | NMOSFET: 1 | Voltage source: 1 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| V1   | VDC          | 12 V            | +=VDD; -=GND           |
| R1   | RES          | 100 kOhm        | 1=VDD; 2=GATE          |
| C1   | CAP          | 10 uF           | 1=GATE; 2=GND          |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |
| R2   | RES          | 1 kOhm          | 1=VDD; 2=LED_A         |
| LED1 | LED          | RED             | A=LED_A; K=DRAIN       |

### Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | V1.-, C1.2, Q1.S          |
| VDD   | V1.+ , R1.1, R2.1         |
| DRAIN | Q1.D, LED1.K              |
| GATE  | R1.2, C1.1, Q1.G          |
| LED_A | R2.2, LED1.A              |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 14: RC Delay-Off Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | Diode: 1 | LED: 1 | NMOSFET: 1 | Voltage source: 1 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net       |
|------|--------------|---------------------|------------------------|
| V1   | VPULSE       | 0-12 V, slow on/off | +=VDD; -=GND           |
| R1   | RES          | 100 kOhm            | 1=VDD; 2=GATE          |
| C1   | CAP          | 10 uF               | 1=GATE; 2=GND          |
| D1   | DIODE        | 1N4148              | A=VDD; K=GATE          |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=DRAIN; S=GND |
| R2   | RES          | 1 kOhm              | 1=VDD; 2=LED_A         |
| LED1 | LED          | RED                 | A=LED_A; K=DRAIN       |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | V1.-, C1.2, Q1.S          |
| VDD   | V1.+, R1.1, D1.A, R2.1    |
| DRAIN | Q1.D, LED1.K              |
| GATE  | R1.2, C1.1, D1.K, Q1.G    |
| LED_A | R2.2, LED1.A              |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 15: First-Order RC Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 1.59 kOhm       | 1=VIN; 2=VOUT    |
| C1  | CAP          | 100 nF          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, C1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 16: First-Order RC High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| C1  | CAP          | 100 nF          | 1=VIN; 2=VOUT    |
| R1  | RES          | 1.59 kOhm       | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2                |
| VIN  | V1.+, C1.1                |
| VOUT | C1.2, R1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 17: Two-Stage Passive RC Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 2 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=N1      |
| C1  | CAP          | 100 nF          | 1=N1; 2=GND      |
| R2  | RES          | 1 kOhm          | 1=N1; 2=VOUT     |
| C2  | CAP          | 100 nF          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2, C2.2          |
| VIN  | V1.+ , R1.1               |
| N1   | R1.2, C1.1, R2.1          |
| VOUT | R2.2, C2.1                |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 18: Two-Stage Passive RC High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 2 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| C1  | CAP          | 100 nF          | 1=VIN; 2=N1      |
| R1  | RES          | 1 kOhm          | 1=N1; 2=GND      |
| C2  | CAP          | 100 nF          | 1=N1; 2=VOUT     |
| R2  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2, R2.2          |
| VIN  | V1.+, C1.1                |
| N1   | C1.2, R1.1, C2.1          |
| VOUT | C2.2, R2.1                |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 19: Passive RC Band-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 2 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| C1  | CAP          | 100 nF          | 1=VIN; 2=N1      |
| R1  | RES          | 10 kOhm         | 1=N1; 2=GND      |
| R2  | RES          | 1 kOhm          | 1=N1; 2=VOUT     |
| C2  | CAP          | 10 nF           | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2, C2.2          |
| VIN  | V1.+, C1.1                |
| N1   | C1.2, R1.1, R2.1          |
| VOUT | R2.2, C2.1                |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 20: Twin-T RC Notch Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 3 | Voltage source: 1 | Total: 7

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm         | 1=VIN; 2=NRA     |
| R2  | RES          | 10 kOhm         | 1=NRA; 2=VOUT    |
| C3  | CAP          | 20 nF           | 1=NRA; 2=GND     |
| C1  | CAP          | 10 nF           | 1=VIN; 2=NCA     |
| C2  | CAP          | 10 nF           | 1=NCA; 2=VOUT    |
| R3  | RES          | 5 kOhm          | 1=NCA; 2=GND     |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C3.2, R3.2          |
| VIN  | V1.+, R1.1, C1.1          |
| NCA  | C1.2, C2.1, R3.1          |
| NRA  | R1.2, R2.1, C3.1          |
| VOUT | R2.2, C2.2                |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 21: Bridged-T RC Notch Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 2 | Voltage source: 1 | Total: 7

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm         | 1=VIN; 2=N1      |
| R2  | RES          | 10 kOhm         | 1=N1; 2=VOUT     |
| R3  | RES          | 20 kOhm         | 1=VIN; 2=VOUT    |
| R4  | RES          | 5 kOhm          | 1=N1; 2=GND      |
| C1  | CAP          | 10 nF           | 1=VIN; 2=N1      |
| C2  | CAP          | 10 nF           | 1=N1; 2=VOUT     |

## Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | V1.-, R4.2                   |
| VIN  | V1.+, R1.1, R3.1, C1.1       |
| N1   | R1.2, R2.1, R4.1, C1.2, C2.1 |
| VOUT | R2.2, R3.2, C2.2             |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 22: RC Phase-Shift Network

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 3 | Voltage source: 1 | Total: 7

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| C1  | CAP          | 10 nF           | 1=VIN; 2=N1      |
| R1  | RES          | 10 kOhm         | 1=N1; 2=GND      |
| C2  | CAP          | 10 nF           | 1=N1; 2=N2       |
| R2  | RES          | 10 kOhm         | 1=N2; 2=GND      |
| C3  | CAP          | 10 nF           | 1=N2; 2=VOUT     |
| R3  | RES          | 10 kOhm         | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2, R2.2, R3.2    |
| VIN  | V1.+ , C1.1               |
| N1   | C1.2, R1.1, C2.1          |
| N2   | C2.2, R2.1, C3.1          |
| VOUT | C3.2, R3.1                |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 23: Passive RC Integrator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 0-5 V, 10 kHz   | +=VIN; -=GND     |
| R1  | RES          | 10 kOhm         | 1=VIN; 2=VOUT    |
| C1  | CAP          | 10 nF           | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, C1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 24: Passive RC Differentiator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 0-5 V, 1 kHz    | +=VIN; -=GND     |
| C1  | CAP          | 10 nF           | 1=VIN; 2=VOUT    |
| R1  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2                |
| VIN  | V1.+, C1.1                |
| VOUT | C1.2, R1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 25: Capacitive Voltage Divider

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Capacitor: 2 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| C1  | CAP          | 100 nF          | 1=VIN; 2=VOUT    |
| C2  | CAP          | 100 nF          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C2.2                |
| VIN  | V1.+, C1.1                |
| VOUT | C1.2, C2.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 26: Capacitor Reservoir Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 1 | Diode: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC; -=GND      |
| D1  | DIODE        | 1N4007          | A=AC; K=VOUT     |
| C1  | CAP-ELEC     | 1000 uF         | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.-, RL.2          |
| AC   | V1.+, D1.A                |
| VOUT | D1.K, C1.+, RL.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 27: Supply Bypass Capacitor Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 5 V             | +=VDD; -=GND     |
| RL  | RES          | 1 kOhm          | 1=VDD; 2=GND     |
| C1  | CAP          | 100 nF          | 1=VDD; 2=GND     |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, RL.2, C1.2          |
| VDD | V1.+, RL.1, C1.1          |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 28: Power-Supply Decoupling Network

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | Electrolytic capacitor: 1 | Voltage source: 1 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| V1   | VDC          | 12 V            | +=VIN; -=GND     |
| RISO | RES          | 10 Ohm          | 1=VIN; 2=VDD     |
| RL   | RES          | 1 kOhm          | 1=VDD; 2=GND     |
| C1   | CAP          | 100 nF          | 1=VDD; 2=GND     |
| C2   | CAP-ELEC     | 100 uF          | +=VDD; -=GND     |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, RL.2, C1.2, C2.-    |
| VDD | RISO.2, RL.1, C1.1, C2.+  |
| VIN | V1.+, RISO.1              |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 29: Electrolytic Smoothing Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 1 | Diode: 4 | Voltage source: 1 | Total: 7

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=VOUT    |
| D2  | DIODE        | 1N4007          | A=AC2; K=VOUT    |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| C1  | CAP-ELEC     | 1000 uF         | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | D3.A, D4.A, C1.-, RL.2    |
| AC1  | V1.+, D1.A, D3.K          |
| AC2  | V1.-, D2.A, D4.K          |
| VOUT | D1.K, D2.K, C1.+, RL.1    |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 30: Capacitor Bleeder-Discharge Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VOUT; -=GND    |
| C1  | CAP-ELEC     | 1000 uF         | +=VOUT; -=GND    |
| RB  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.-, RB.2          |
| VOUT | V1.+, C1.+, RB.1          |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 31: RL Charging Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Inductor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 0-10 V, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 100 Ohm         | 1=VIN; 2=VOUT    |
| L1  | INDUCTOR     | 10 mH           | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, L1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, L1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 32: RL Discharging Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Inductor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 10-0 V, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 100 Ohm         | 1=VIN; 2=VOUT    |
| L1  | INDUCTOR     | 10 mH           | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, L1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, L1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 33: First-Order RL Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Inductor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| L1  | INDUCTOR     | 10 mH           | 1=VIN; 2=VOUT    |
| R1  | RES          | 100 Ohm         | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2                |
| VIN  | V1.+, L1.1                |
| VOUT | L1.2, R1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 34: First-Order RL High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Inductor: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 100 Ohm         | 1=VIN; 2=VOUT    |
| L1  | INDUCTOR     | 10 mH           | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, L1.2                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, L1.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 35: Series RLC Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 100 Ohm         | 1=VIN; 2=N1      |
| L1  | INDUCTOR     | 10 mH           | 1=N1; 2=N2       |
| C1  | CAP          | 2.2 uF          | 1=N2; 2=GND      |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, C1.2                |
| VIN | V1.+, R1.1                |
| N1  | R1.2, L1.1                |
| N2  | L1.2, C1.1                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 36: Parallel RLC Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=GND     |
| L1  | INDUCTOR     | 10 mH           | 1=VIN; 2=GND     |
| C1  | CAP          | 2.2 uF          | 1=VIN; 2=GND     |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, R1.2, L1.2, C1.2    |
| VIN | V1.+, R1.1, L1.1, C1.1    |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 37: Series RLC Resonant Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| R1  | RES          | 100 Ohm         | 1=VIN; 2=VOUT    |
| L1  | INDUCTOR     | 10 mH           | 1=VOUT; 2=N2     |
| C1  | CAP          | 2.53 uF         | 1=N2; 2=GND      |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| N2   | L1.2, C1.1                |
| VOUT | R1.2, L1.1                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 38: Parallel RLC Resonant Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=GND     |
| L1  | INDUCTOR     | 10 mH           | 1=VIN; 2=GND     |
| C1  | CAP          | 2.53 uF         | 1=VIN; 2=GND     |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, R1.2, L1.2, C1.2    |
| VIN | V1.+, R1.1, L1.1, C1.1    |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 39: Series RLC Band-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| L1  | INDUCTOR     | 10 mH           | 1=VIN; 2=N1      |
| C1  | CAP          | 2.53 uF         | 1=N1; 2=VOUT     |
| R1  | RES          | 100 Ohm         | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, R1.2                |
| VIN  | V1.+, L1.1                |
| N1   | L1.2, C1.1                |
| VOUT | C1.2, R1.1                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 40: Parallel RLC Band-Stop Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| R1  | RES          | 100 Ohm         | 1=VIN; 2=VOUT    |
| L1  | INDUCTOR     | 10 mH           | 1=VOUT; 2=N1     |
| C1  | CAP          | 2.53 uF         | 1=VOUT; 2=N1     |
| RL  | RES          | 1 kOhm          | 1=N1; 2=GND      |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, R1.1                |
| N1   | L1.2, C1.2, RL.1          |
| VOUT | R1.2, L1.1, C1.1          |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 41: LC Series Resonant Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| R1  | RES          | 10 Ohm          | 1=VIN; 2=N1      |
| L1  | INDUCTOR     | 10 mH           | 1=N1; 2=N2       |
| C1  | CAP          | 2.53 uF         | 1=N2; 2=GND      |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, C1.2                |
| VIN | V1.+, R1.1                |
| N1  | R1.2, L1.1                |
| N2  | L1.2, C1.1                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 42: LC Parallel Tank Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Capacitor: 1 | Inductor: 1 | Current source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| I1  | IDC          | 1 mA AC         | +=TANK; -=GND    |
| L1  | INDUCTOR     | 10 mH           | 1=TANK; 2=GND    |
| C1  | CAP          | 2.53 uF         | 1=TANK; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | I1.-, L1.2, C1.2          |
| TANK | I1.+, L1.1, C1.1          |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 43: L-Section LC Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| L1  | INDUCTOR     | 10 mH           | 1=VIN; 2=VOUT    |
| C1  | CAP          | 1 uF            | 1=VOUT; 2=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2, RL.2          |
| VIN  | V1.+, L1.1                |
| VOUT | L1.2, C1.1, RL.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 44: L-Section LC High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| C1  | CAP          | 1 uF            | 1=VIN; 2=VOUT    |
| L1  | INDUCTOR     | 10 mH           | 1=VOUT; 2=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, L1.2, RL.2          |
| VIN  | V1.+, C1.1                |
| VOUT | C1.2, L1.1, RL.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 45: Pi-Section LC Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 2 | Inductor: 1 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| C1  | CAP          | 1 uF            | 1=VIN; 2=GND     |
| L1  | INDUCTOR     | 10 mH           | 1=VIN; 2=VOUT    |
| C2  | CAP          | 1 uF            | 1=VOUT; 2=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2, C2.2, RL.2    |
| VIN  | V1.+, C1.1, L1.1          |
| VOUT | L1.2, C2.1, RL.1          |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 46: T-Section LC Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 2 | Voltage source: 1 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 1 Vpk, sweep    | +=VIN; -=GND     |
| L1  | INDUCTOR     | 5 mH            | 1=VIN; 2=N1      |
| C1  | CAP          | 2 uF            | 1=N1; 2=GND      |
| L2  | INDUCTOR     | 5 mH            | 1=N1; 2=VOUT     |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2, RL.2          |
| VIN  | V1.+, L1.1                |
| N1   | L1.2, C1.1, L2.1          |
| VOUT | L2.2, RL.1                |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 47: CLC Pi Power-Supply Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 2 | Inductor: 1 | Diode: 4 | Voltage source: 1 | Total: 9

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=RAW     |
| D2  | DIODE        | 1N4007          | A=AC2; K=RAW     |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| C1  | CAP-ELEC     | 1000 uF         | +=RAW; -=GND     |
| L1  | INDUCTOR     | 100 mH          | 1=RAW; 2=VOUT    |
| C2  | CAP-ELEC     | 470 uF          | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | D3.A, D4.A, C1.-, C2.-, RL.2 |
| AC1  | V1.+, D1.A, D3.K             |
| AC2  | V1.-, D2.A, D4.K             |
| RAW  | D1.K, D2.K, C1.+, L1.1       |
| VOUT | L1.2, C2.+, RL.1             |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 48: CRC Pi Power-Supply Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Electrolytic capacitor: 2 | Diode: 4 | Voltage source: 1 | Total: 9

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=RAW     |
| D2  | DIODE        | 1N4007          | A=AC2; K=RAW     |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| C1  | CAP-ELEC     | 1000 uF         | +=RAW; -=GND     |
| R1  | RES          | 100 Ohm         | 1=RAW; 2=VOUT    |
| C2  | CAP-ELEC     | 470 uF          | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | D3.A, D4.A, C1.-, C2.-, RL.2 |
| AC1  | V1.+, D1.A, D3.K             |
| AC2  | V1.-, D2.A, D4.K             |
| RAW  | D1.K, D2.K, C1.+, R1.1       |
| VOUT | R1.2, C2.+, RL.1             |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 49: RLC Damping Network

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 0-10 V step     | +=VIN; -=GND     |
| R1  | RES          | 47 Ohm          | 1=VIN; 2=N1      |
| L1  | INDUCTOR     | 10 mH           | 1=N1; 2=VOUT     |
| C1  | CAP          | 1 uF            | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| N1   | R1.2, L1.1                |
| VOUT | L1.2, C1.1                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 50: Damped LC Step-Response Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Inductor: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 0-10 V step     | +=VIN; -=GND     |
| R1  | RES          | 22 Ohm          | 1=VIN; 2=N1      |
| L1  | INDUCTOR     | 10 mH           | 1=N1; 2=VOUT     |
| C1  | CAP          | 1 uF            | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2                |
| VIN  | V1.+, R1.1                |
| N1   | R1.2, L1.1                |
| VOUT | L1.2, C1.1                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 51: Half-Wave Rectifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC; -=GND      |
| D1  | DIODE        | 1N4007          | A=AC; K=VOUT     |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| AC   | V1.+, D1.A                |
| VOUT | D1.K, RL.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 52: Full-Wave Bridge Rectifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 4 | Voltage source: 1 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=VOUT    |
| D2  | DIODE        | 1N4007          | A=AC2; K=VOUT    |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | D3.A, D4.A, RL.2          |
| AC1  | V1.+, D1.A, D3.K          |
| AC2  | V1.-, D2.A, D4.K          |
| VOUT | D1.K, D2.K, RL.1          |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 53: Half-Wave Rectifier with Capacitor Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 1 | Diode: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC; -=GND      |
| D1  | DIODE        | 1N4007          | A=AC; K=VOUT     |
| C1  | CAP-ELEC     | 1000 uF         | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.-, RL.2          |
| AC   | V1.+, D1.A                |
| VOUT | D1.K, C1.+, RL.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 54: Bridge Rectifier with Capacitor Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 1 | Diode: 4 | Voltage source: 1 | Total: 7

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=VOUT    |
| D2  | DIODE        | 1N4007          | A=AC2; K=VOUT    |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| C1  | CAP-ELEC     | 1000 uF         | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | D3.A, D4.A, C1.-, RL.2    |
| AC1  | V1.+, D1.A, D3.K          |
| AC2  | V1.-, D2.A, D4.K          |
| VOUT | D1.K, D2.K, C1.+, RL.1    |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 55: Bridge Rectifier with CRC Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Electrolytic capacitor: 2 | Diode: 4 | Voltage source: 1 | Total: 9

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=RAW     |
| D2  | DIODE        | 1N4007          | A=AC2; K=RAW     |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| C1  | CAP-ELEC     | 1000 uF         | +=RAW; -=GND     |
| R1  | RES          | 100 Ohm         | 1=RAW; 2=VOUT    |
| C2  | CAP-ELEC     | 470 uF          | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | D3.A, D4.A, C1.-, C2.-, RL.2 |
| AC1  | V1.+, D1.A, D3.K             |
| AC2  | V1.-, D2.A, D4.K             |
| RAW  | D1.K, D2.K, C1.+, R1.1       |
| VOUT | R1.2, C2.+, RL.1             |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 56: Bridge Rectifier with CLC Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 2 | Inductor: 1 | Diode: 4 | Voltage source: 1 | Total: 9

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 12 Vrms, 50 Hz  | +=AC1; -=AC2     |
| D1  | DIODE        | 1N4007          | A=AC1; K=RAW     |
| D2  | DIODE        | 1N4007          | A=AC2; K=RAW     |
| D3  | DIODE        | 1N4007          | A=GND; K=AC1     |
| D4  | DIODE        | 1N4007          | A=GND; K=AC2     |
| C1  | CAP-ELEC     | 1000 uF         | +=RAW; -=GND     |
| L1  | INDUCTOR     | 100 mH          | 1=RAW; 2=VOUT    |
| C2  | CAP-ELEC     | 470 uF          | +=VOUT; -=GND    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | D3.A, D4.A, C1.-, C2.-, RL.2 |
| AC1  | V1.+, D1.A, D3.K             |
| AC2  | V1.-, D2.A, D4.K             |
| RAW  | D1.K, D2.K, C1.+, L1.1       |
| VOUT | L1.2, C2.+, RL.1             |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 57: Positive Series Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| D1  | DIODE        | 1N4148          | A=VOUT; K=VIN    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, D1.K                |
| VOUT | D1.A, RL.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 58: Negative Series Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| D1  | DIODE        | 1N4148          | A=VIN; K=VOUT    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, D1.A                |
| VOUT | D1.K, RL.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 59: Positive Shunt Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| D1  | DIODE        | 1N4148          | A=VOUT; K=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.K                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, D1.A                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 60: Negative Shunt Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| D1  | DIODE        | 1N4148          | A=GND; K=VOUT    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.A                |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, D1.K                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 61: Symmetrical Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 2 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| D1  | DIODE        | 1N4148          | A=VOUT; K=GND    |
| D2  | DIODE        | 1N4148          | A=GND; K=VOUT    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.K, D2.A          |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, D1.A, D2.K          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 62: Biased Positive Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 2 | Total: 4

| Ref   | Proteus part | Value / setting | Every pin -> net |
|-------|--------------|-----------------|------------------|
| V1    | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| VBIAS | VDC          | 3 V             | +=BIAS; -=GND    |
| R1    | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| D1    | DIODE        | 1N4148          | A=VOUT; K=BIAS   |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, VBIAS.-             |
| VIN  | V1.+, R1.1                |
| BIAS | VBIAS.+, D1.K             |
| VOUT | R1.2, D1.A                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 63: Biased Negative Diode Clipper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 2 | Total: 4

| Ref   | Proteus part | Value / setting | Every pin -> net |
|-------|--------------|-----------------|------------------|
| V1    | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| VBIAS | VDC          | 3 V             | +=GND; -=BIAS    |
| R1    | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| D1    | DIODE        | 1N4148          | A=BIAS; K=VOUT   |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, VBIAS.+             |
| VIN  | V1.+, R1.1                |
| BIAS | VBIAS.-, D1.A             |
| VOUT | R1.2, D1.K                |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 64: Positive Diode Clamper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Diode: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| C1  | CAP          | 1 uF            | 1=VIN; 2=VOUT    |
| D1  | DIODE        | 1N4148          | A=GND; K=VOUT    |
| R1  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.A, R1.2          |
| VIN  | V1.+, C1.1                |
| VOUT | C1.2, D1.K, R1.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 65: Negative Diode Clamper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Diode: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| C1  | CAP          | 1 uF            | 1=VIN; 2=VOUT    |
| D1  | DIODE        | 1N4148          | A=VOUT; K=GND    |
| R1  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.K, R1.2          |
| VIN  | V1.+, C1.1                |
| VOUT | C1.2, D1.A, R1.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 66: Biased Diode Clamper

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Diode: 1 | Voltage source: 2 | Total: 5

| Ref   | Proteus part | Value / setting | Every pin -> net |
|-------|--------------|-----------------|------------------|
| V1    | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| VBIAS | VDC          | 2 V             | +=BIAS; -=GND    |
| C1    | CAP          | 1 uF            | 1=VIN; 2=VOUT    |
| D1    | DIODE        | 1N4148          | A=BIAS; K=VOUT   |
| R1    | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, VBIAS.-, R1.2       |
| VIN  | V1.+, C1.1                |
| BIAS | VBIAS.+, D1.A             |
| VOUT | C1.2, D1.K, R1.1          |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 67: Diode Peak Detector

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Diode: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=VIN; -=GND     |
| D1  | DIODE        | 1N4148          | A=VIN; K=VOUT    |
| C1  | CAP          | 1 uF            | 1=VOUT; 2=GND    |
| RL  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2, RL.2          |
| VIN  | V1.+, D1.A                |
| VOUT | D1.K, C1.1, RL.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 68: Diode Envelope Detector

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Diode: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | AM signal       | +=VIN; -=GND     |
| D1  | DIODE        | 1N4148          | A=VIN; K=VOUT    |
| C1  | CAP          | 10 nF           | 1=VOUT; 2=GND    |
| R1  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, C1.2, R1.2          |
| VIN  | V1.+, D1.A                |
| VOUT | D1.K, C1.1, R1.1          |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 69: Diode Charge-Pump Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 2 | Diode: 2 | Voltage source: 1 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VPULSE       | 0-5 V, 10 kHz   | +=CLK; -=GND     |
| C1  | CAP          | 1 uF            | 1=CLK; 2=N1      |
| D1  | DIODE        | 1N4148          | A=GND; K=N1      |
| D2  | DIODE        | 1N4148          | A=N1; K=VOUT     |
| C2  | CAP          | 10 uF           | 1=VOUT; 2=GND    |
| RL  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.A, C2.2, RL.2    |
| CLK  | V1.+, C1.1                |
| N1   | C1.2, D1.K, D2.A          |
| VOUT | D2.K, C2.1, RL.1          |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 70: Half-Wave Voltage Doubler

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 2 | Diode: 2 | Voltage source: 1 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=AC; -=GND      |
| C1  | CAP          | 10 uF           | 1=AC; 2=N1       |
| D1  | DIODE        | 1N4148          | A=GND; K=N1      |
| D2  | DIODE        | 1N4148          | A=N1; K=VOUT     |
| C2  | CAP          | 10 uF           | 1=VOUT; 2=GND    |
| RL  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.A, C2.2, RL.2    |
| AC   | V1.+, C1.1                |
| N1   | C1.2, D1.K, D2.A          |
| VOUT | D2.K, C2.1, RL.1          |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 71: Full-Wave Voltage Doubler

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Electrolytic capacitor: 2 | Diode: 2 | Voltage source: 1 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=AC; -=GND      |
| D1  | DIODE        | 1N4148          | A=AC; K=VP       |
| D2  | DIODE        | 1N4148          | A=GND; K=VN      |
| C1  | CAP-ELEC     | 10 uF           | +=VP; -=GND      |
| C2  | CAP-ELEC     | 10 uF           | +=GND; -=VN      |
| RL  | RES          | 100 kOhm        | 1=VP; 2=VN       |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, D2.A, C1.-, C2.+    |
| AC  | V1.+, D1.A                |
| VN  | D2.K, C2.-, RL.2          |
| VP  | D1.K, C1.+, RL.1          |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 72: Diode Voltage Tripler

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 3 | Diode: 3 | Voltage source: 1 | Total: 8

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=AC; -=GND      |
| C1  | CAP          | 10 uF           | 1=AC; 2=N1       |
| D1  | DIODE        | 1N4148          | A=GND; K=N1      |
| D2  | DIODE        | 1N4148          | A=N1; K=N2       |
| C2  | CAP          | 10 uF           | 1=N2; 2=GND      |
| C3  | CAP          | 10 uF           | 1=N2; 2=VOUT     |
| D3  | DIODE        | 1N4148          | A=AC; K=VOUT     |
| RL  | RES          | 100 kOhm        | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.A, C2.2, RL.2    |
| AC   | V1.+, C1.1, D3.A          |
| N1   | C1.2, D1.K, D2.A          |
| N2   | D2.K, C2.1, C3.1          |
| VOUT | C3.2, D3.K, RL.1          |

PIN AUDIT: PASS - 16 of 16 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 73: Four-Stage Cockcroft-Walton Multiplier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 8 | Diode: 8 | Voltage source: 1 | Total: 18

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 10 Vpk, 1 kHz   | +=AC; -=GND      |
| C1  | CAP          | 1 uF            | 1=AC; 2=N1       |
| D1  | DIODE        | 1N4007          | A=GND; K=N1      |
| D2  | DIODE        | 1N4007          | A=N1; K=N2       |
| C2  | CAP          | 1 uF            | 1=N2; 2=GND      |
| C3  | CAP          | 1 uF            | 1=N1; 2=N3       |
| D3  | DIODE        | 1N4007          | A=N2; K=N3       |
| D4  | DIODE        | 1N4007          | A=N3; K=N4       |
| C4  | CAP          | 1 uF            | 1=N4; 2=N2       |
| C5  | CAP          | 1 uF            | 1=N3; 2=N5       |
| D5  | DIODE        | 1N4007          | A=N4; K=N5       |
| D6  | DIODE        | 1N4007          | A=N5; K=N6       |
| C6  | CAP          | 1 uF            | 1=N6; 2=N4       |
| C7  | CAP          | 1 uF            | 1=N5; 2=N7       |
| D7  | DIODE        | 1N4007          | A=N6; K=N7       |
| D8  | DIODE        | 1N4007          | A=N7; K=VOUT     |
| C8  | CAP          | 1 uF            | 1=VOUT; 2=N6     |
| RL  | RES          | 10 MOhm         | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, D1.A, C2.2, RL.2    |
| AC   | V1.+, C1.1                |
| N1   | C1.2, D1.K, D2.A, C3.1    |
| N2   | D2.K, C2.1, D3.A, C4.2    |
| N3   | C3.2, D3.K, D4.A, C5.1    |
| N4   | D4.K, C4.1, D5.A, C6.2    |
| N5   | C5.2, D5.K, D6.A, C7.1    |
| N6   | D6.K, C6.1, D7.A, C8.2    |
| N7   | C7.2, D7.K, D8.A          |
| VOUT | D8.K, C8.1, RL.1          |

PIN AUDIT: PASS - 36 of 36 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 74: Diode OR Gate

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 2 | Voltage source: 2 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| VA  | VDC          | 5 V             | +=A; -=GND       |
| VB  | VDC          | 0 V             | +=B; -=GND       |
| D1  | DIODE        | 1N4148          | A=A; K=Y         |
| D2  | DIODE        | 1N4148          | A=B; K=Y         |
| R1  | RES          | 10 kOhm         | 1=Y; 2=GND       |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | VA.-, VB.-, R1.2          |
| A   | VA.+ , D1.A               |
| B   | VB.+ , D2.A               |
| Y   | D1.K, D2.K, R1.1          |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 75: Diode AND Gate

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 2 | Voltage source: 3 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| VA  | VDC          | 5 V             | +=A; -=GND       |
| VB  | VDC          | 5 V             | +=B; -=GND       |
| VCC | VDC          | 5 V             | +=VDD; -=GND     |
| R1  | RES          | 10 kOhm         | 1=VDD; 2=Y       |
| D1  | DIODE        | 1N4148          | A=Y; K=A         |
| D2  | DIODE        | 1N4148          | A=Y; K=B         |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | VA.-, VB.-, VCC.-         |
| VDD | VCC.+, R1.1               |
| A   | VA.+, D1.K                |
| B   | VB.+, D2.K                |
| Y   | R1.2, D1.A, D2.A          |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 76: Diode Polarity-Selector Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 2 | Voltage source: 2 | Total: 5

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=IN1; -=GND     |
| V2  | VDC          | 9 V             | +=IN2; -=GND     |
| D1  | DIODE        | 1N4007          | A=IN1; K=VOUT    |
| D2  | DIODE        | 1N4007          | A=IN2; K=VOUT    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, V2.-, RL.2          |
| IN1  | V1.+, D1.A                |
| IN2  | V2.+, D2.A                |
| VOUT | D1.K, D2.K, RL.1          |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 77: Bridge Polarity-Correction Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 4 | Voltage source: 1 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V reversible | +=IN1; -=IN2     |
| D1  | DIODE        | 1N4007          | A=IN1; K=VOUT    |
| D2  | DIODE        | 1N4007          | A=IN2; K=VOUT    |
| D3  | DIODE        | 1N4007          | A=GND; K=IN1     |
| D4  | DIODE        | 1N4007          | A=GND; K=IN2     |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

### Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | D3.A, D4.A, RL.2          |
| IN1  | V1.+, D1.A, D3.K          |
| IN2  | V1.-, D2.A, D4.K          |
| VOUT | D1.K, D2.K, RL.1          |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 78: Series-Diode Reverse-Polarity Protection

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| D1  | DIODE        | 1N4007          | A=VIN; K=VOUT    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, D1.A                |
| VOUT | D1.K, RL.1                |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 79: Freewheel Diode Clamp

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref   | Proteus part | Value / setting | Every pin -> net       |
|-------|--------------|-----------------|------------------------|
| V1    | VDC          | 12 V            | +=VDD; -=GND           |
| VG    | VPULSE       | 0-5 V           | +=GATE; -=GND          |
| RCOIL | RES          | 100 Ohm         | 1=VDD; 2=N1            |
| L1    | INDUCTOR     | 100 mH          | 1=N1; 2=DRAIN          |
| D1    | DIODE        | 1N4007          | A=DRAIN; K=VDD         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | V1.-, VG.-, Q1.S          |
| VDD   | V1.+, RCOIL.1, D1.K       |
| DRAIN | L1.2, D1.A, Q1.D          |
| GATE  | VG.+, Q1.G                |
| N1    | RCOIL.2, L1.1             |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

## Circuit 80: NMOS Inductive-Load Switch with Flyback Diode

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref | Proteus part | Value / setting | Every pin -> net       |
|-----|--------------|-----------------|------------------------|
| V1  | VDC          | 12 V            | +=VDD; -=GND           |
| VG  | VPULSE       | 0-5 V           | +=GDRV; -=GND          |
| RG  | RES          | 100 Ohm         | 1=GDRV; 2=GATE         |
| L1  | INDUCTOR     | 100 mH          | 1=VDD; 2=DRAIN         |
| D1  | DIODE        | 1N4007          | A=DRAIN; K=VDD         |
| Q1  | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

### Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | V1.-, VG.-, Q1.S          |
| VDD   | V1.+, L1.1, D1.K          |
| DRAIN | L1.2, D1.A, Q1.D          |
| GATE  | RG.2, Q1.G                |
| GDRV  | VG.+, RG.1                |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 81: Diode-RC Snubber for an Inductive Load

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 8

| Ref | Proteus part | Value / setting | Every pin -> net       |
|-----|--------------|-----------------|------------------------|
| V1  | VDC          | 12 V            | +=VDD; -=GND           |
| VG  | VPULSE       | 0-5 V           | +=GDRV; -=GND          |
| RG  | RES          | 100 Ohm         | 1=GDRV; 2=GATE         |
| L1  | INDUCTOR     | 100 mH          | 1=VDD; 2=DRAIN         |
| D1  | DIODE        | 1N4148          | A=DRAIN; K=SNUB        |
| RS  | RES          | 100 Ohm         | 1=SNUB; 2=CSN          |
| CS  | CAP          | 100 nF          | 1=CSN; 2=VDD           |
| Q1  | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | V1.-, VG.-, Q1.S          |
| VDD   | V1.+, L1.1, CS.2          |
| CSN   | RS.2, CS.1                |
| DRAIN | L1.2, D1.A, Q1.D          |
| GATE  | RG.2, Q1.G                |
| GDRV  | VG.+, RG.1                |
| SNUB  | D1.K, RS.1                |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 82: Zener Shunt Voltage Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Zener: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 330 Ohm         | 1=VIN; 2=VOUT    |
| DZ1 | ZENER        | 5.1 V           | A=GND; K=VOUT    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, DZ1.A, RL.2         |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, DZ1.K, RL.1         |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 83: Zener Voltage-Reference Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Zener: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 680 Ohm         | 1=VIN; 2=VREF    |
| DZ1 | ZENER        | 5.1 V           | A=GND; K=VREF    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, DZ1.A               |
| VIN  | V1.+, R1.1                |
| VREF | R1.2, DZ1.K               |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 84: Zener Overvoltage Clamp

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Zener: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 0-12 V sweep    | +=VIN; -=GND     |
| R1  | RES          | 330 Ohm         | 1=VIN; 2=VOUT    |
| DZ1 | ZENER        | 5.1 V           | A=GND; K=VOUT    |
| RL  | RES          | 1 kOhm          | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, DZ1.A, RL.2         |
| VIN  | V1.+, R1.1                |
| VOUT | R1.2, DZ1.K, RL.1         |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 85: Back-to-Back Zener AC Limiter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Zener: 2 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VSINE        | 20 Vpk, 1 kHz   | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=VOUT    |
| DZ1 | ZENER        | 5.1 V           | A=MID; K=VOUT    |
| DZ2 | ZENER        | 5.1 V           | A=MID; K=GND     |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, DZ2.K               |
| VIN  | V1.+, R1.1                |
| MID  | DZ1.A, DZ2.A              |
| VOUT | R1.2, DZ1.K               |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 86: Zener Level-Shifter Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Zener: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net |
|-----|--------------|-----------------|------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND     |
| R1  | RES          | 1 kOhm          | 1=VIN; 2=N1      |
| DZ1 | ZENER        | 5.1 V           | A=VOUT; K=N1     |
| RL  | RES          | 10 kOhm         | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, R1.1                |
| N1   | R1.2, DZ1.K               |
| VOUT | DZ1.A, RL.1               |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 87: Resistor-Limited LED Indicator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 1 | Voltage source: 1 | Total: 3

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| V1   | VDC          | 5 V             | +=VDD; -=GND     |
| R1   | RES          | 330 Ohm         | 1=VDD; 2=LED_A   |
| LED1 | LED          | RED             | A=LED_A; K=GND   |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | V1.-, LED1.K              |
| VDD   | V1.+, R1.1                |
| LED_A | R1.2, LED1.A              |

PIN AUDIT: PASS - 6 of 6 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 88: Anti-Parallel LED Polarity Indicator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 2 | Voltage source: 1 | Total: 4

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| V1   | VSINE        | 5 Vpk, 1 Hz     | +=VIN; -=GND     |
| R1   | RES          | 330 Ohm         | 1=VIN; 2=N1      |
| LED1 | LED          | RED             | A=N1; K=GND      |
| LED2 | LED          | GREEN           | A=GND; K=N1      |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, LED1.K, LED2.A      |
| VIN | V1.+, R1.1                |
| N1  | R1.2, LED1.A, LED2.K      |

PIN AUDIT: PASS - 8 of 8 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 89: Series LED String

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 3 | Voltage source: 1 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| V1   | VDC          | 12 V            | +=VDD; -=GND     |
| R1   | RES          | 330 Ohm         | 1=VDD; 2=N1      |
| LED1 | LED          | RED             | A=N1; K=N2       |
| LED2 | LED          | GREEN           | A=N2; K=N3       |
| LED3 | LED          | BLUE            | A=N3; K=GND      |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | V1.-, LED3.K              |
| VDD | V1.+, R1.1                |
| N1  | R1.2, LED1.A              |
| N2  | LED1.K, LED2.A            |
| N3  | LED2.K, LED3.A            |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 90: Parallel LED Branches with Ballast Resistors

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | LED: 3 | Voltage source: 1 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| V1   | VDC          | 5 V             | +=VDD; -=GND     |
| R1   | RES          | 330 Ohm         | 1=VDD; 2=A1      |
| LED1 | LED          | RED             | A=A1; K=GND      |
| R2   | RES          | 330 Ohm         | 1=VDD; 2=A2      |
| LED2 | LED          | GREEN           | A=A2; K=GND      |
| R3   | RES          | 330 Ohm         | 1=VDD; 2=A3      |
| LED3 | LED          | BLUE            | A=A3; K=GND      |

## Net connection list

| Net | Pins to join electrically    |
|-----|------------------------------|
| GND | V1.-, LED1.K, LED2.K, LED3.K |
| VDD | V1.+, R1.1, R2.1, R3.1       |
| A1  | R1.2, LED1.A                 |
| A2  | R2.2, LED2.A                 |
| A3  | R3.2, LED3.A                 |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 91: NMOS Low-Side Switch

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 2 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND           |
| VG   | VPULSE       | 0-5 V           | +=GATE; -=GND          |
| RL   | RES          | 1 kOhm          | 1=VDD; 2=DRAIN         |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VG.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RL.1              |
| DRAIN | RL.2, Q1.D                |
| GATE  | VG.+, RPD.1, Q1.G         |

PIN AUDIT: PASS - 11 of 11 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 92: Logic-Level NMOS LED Switch

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | LED: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND           |
| VG   | VPULSE       | 0-5 V           | +=GATE; -=GND          |
| RLED | RES          | 330 Ohm         | 1=VDD; 2=LED_A         |
| LED1 | LED          | RED             | A=LED_A; K=DRAIN       |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VG.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RLED.1            |
| DRAIN | LED1.K, Q1.D              |
| GATE  | VG.+, RPD.1, Q1.G         |
| LED_A | RLED.2, LED1.A            |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 93: NMOS Inductive-Load Driver

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 8

| Ref   | Proteus part | Value / setting | Every pin -> net       |
|-------|--------------|-----------------|------------------------|
| VDD1  | VDC          | 12 V            | +=VDD; -=GND           |
| VG    | VPULSE       | 0-5 V           | +=GDRV; -=GND          |
| RG    | RES          | 100 Ohm         | 1=GDRV; 2=GATE         |
| RPD   | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| RCOIL | RES          | 100 Ohm         | 1=VDD; 2=NCOIL         |
| L1    | INDUCTOR     | 100 mH          | 1=NCOIL; 2=DRAIN       |
| D1    | DIODE        | 1N4007          | A=DRAIN; K=VDD         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VG.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RCOIL.1, D1.K     |
| DRAIN | L1.2, D1.A, Q1.D          |
| GATE  | RG.2, RPD.1, Q1.G         |
| GDRV  | VG.+, RG.1                |
| NCOIL | RCOIL.2, L1.1             |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 94: NMOS Reverse-Polarity Protector

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 1 | Total: 4

| Ref | Proteus part | Value / setting | Every pin -> net     |
|-----|--------------|-----------------|----------------------|
| V1  | VDC          | 12 V            | +=VIN; -=GND         |
| Q1  | NMOSFET      | Generic NMOS    | G=VIN; D=PGND; S=GND |
| RG  | RES          | 100 kOhm        | 1=VIN; 2=GND         |
| RL  | RES          | 1 kOhm          | 1=VIN; 2=PGND        |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, Q1.S, RG.2          |
| VIN  | V1.+, Q1.G, RG.1, RL.1    |
| PGND | Q1.D, RL.2                |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 95: Back-to-Back NMOS Load Disconnect

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 2 | Voltage source: 2 | Total: 6

| Ref   | Proteus part | Value / setting | Every pin -> net      |
|-------|--------------|-----------------|-----------------------|
| VDD1  | VDC          | 12 V            | +=VIN; -=GND          |
| VCTRL | VDC          | 10 V            | +=GATE; -=GND         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=VIN; S=MID  |
| Q2    | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=MID |
| RPD   | RES          | 100 kOhm        | 1=GATE; 2=GND         |
| RL    | RES          | 1 kOhm          | 1=VOUT; 2=GND         |

## Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | VDD1.-, VCTRL.-, RPD.2, RL.2 |
| VIN  | VDD1.+, Q1.D                 |
| GATE | VCTRL.+, Q1.G, Q2.G, RPD.1   |
| MID  | Q1.S, Q2.S                   |
| VOUT | Q2.D, RL.1                   |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 96: NMOS Source Follower

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 4

| Ref  | Proteus part | Value / setting          | Every pin -> net      |
|------|--------------|--------------------------|-----------------------|
| VDD1 | VDC          | 12 V                     | +=VDD; -=GND          |
| VIN1 | VSINE        | 5 V offset, 1 Vpk, 1 kHz | +=GATE; -=GND         |
| Q1   | NMOSFET      | Generic NMOS             | G=GATE; D=VDD; S=VOUT |
| RL   | RES          | 1 kOhm                   | 1=VOUT; 2=GND         |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | VDD1.-, VIN1.-, RL.2      |
| VDD  | VDD1.+, Q1.D              |
| GATE | VIN1.+, Q1.G              |
| VOUT | Q1.S, RL.1                |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 97: NMOS Common-Source Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | NMOSFET: 1 | Voltage source: 2 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net         |
|------|--------------|-----------------|--------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND             |
| VIN1 | VSINE        | 50 mVpk, 1 kHz  | +=GATE; -=GND            |
| R1   | RES          | 1 MOhm          | 1=VDD; 2=GATE            |
| R2   | RES          | 220 kOhm        | 1=GATE; 2=GND            |
| RD   | RES          | 4.7 kOhm        | 1=VDD; 2=VOUT            |
| RS   | RES          | 1 kOhm          | 1=SOURCE; 2=GND          |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=SOURCE |

## Net connection list

| Net    | Pins to join electrically  |
|--------|----------------------------|
| GND    | VDD1.-, VIN1.-, R2.2, RS.2 |
| VDD    | VDD1.+, R1.1, RD.1         |
| GATE   | VIN1.+, R1.2, R2.1, Q1.G   |
| SOURCE | RS.1, Q1.S                 |
| VOUT   | RD.2, Q1.D                 |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 98: Source-Degenerated NMOS Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | NMOSFET: 1 | Voltage source: 2 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net         |
|------|--------------|-----------------|--------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND             |
| VIN1 | VSINE        | 50 mVpk, 1 kHz  | +=GATE; -=GND            |
| R1   | RES          | 1 MOhm          | 1=VDD; 2=GATE            |
| R2   | RES          | 220 kOhm        | 1=GATE; 2=GND            |
| RD   | RES          | 4.7 kOhm        | 1=VDD; 2=VOUT            |
| RS   | RES          | 2.2 kOhm        | 1=SOURCE; 2=GND          |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=SOURCE |

## Net connection list

| Net    | Pins to join electrically  |
|--------|----------------------------|
| GND    | VDD1.-, VIN1.-, R2.2, RS.2 |
| VDD    | VDD1.+, R1.1, RD.1         |
| GATE   | VIN1.+, R1.2, R2.1, Q1.G   |
| SOURCE | RS.1, Q1.S                 |
| VOUT   | RD.2, Q1.D                 |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 99: RC-Coupled NMOS Common-Source Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Capacitor: 2 | NMOSFET: 1 | Voltage source: 2 | Total: 10

| Ref  | Proteus part | Value / setting | Every pin -> net          |
|------|--------------|-----------------|---------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND              |
| VIN1 | VSINE        | 50 mVpk, 1 kHz  | +=VSIG; -=GND             |
| CIN  | CAP          | 1 uF            | 1=VSIG; 2=GATE            |
| R1   | RES          | 1 MOhm          | 1=VDD; 2=GATE             |
| R2   | RES          | 220 kOhm        | 1=GATE; 2=GND             |
| RD   | RES          | 4.7 kOhm        | 1=VDD; 2=DRAIN            |
| RS   | RES          | 1 kOhm          | 1=SOURCE; 2=GND           |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=SOURCE |
| COUT | CAP          | 1 uF            | 1=DRAIN; 2=VOUT           |
| RL   | RES          | 10 kOhm         | 1=VOUT; 2=GND             |

## Net connection list

| Net    | Pins to join electrically        |
|--------|----------------------------------|
| GND    | VDD1.-, VIN1.-, R2.2, RS.2, RL.2 |
| VDD    | VDD1.+, R1.1, RD.1               |
| DRAIN  | RD.2, Q1.D, COUT.1               |
| GATE   | CIN.2, R1.2, R2.1, Q1.G          |
| SOURCE | RS.1, Q1.S                       |
| VSIG   | VIN1.+, CIN.1                    |
| VOUT   | COUT.2, RL.1                     |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 100: NMOS Amplifier with Bypassed Source Resistor

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Capacitor: 3 | NMOSFET: 1 | Voltage source: 2 | Total: 11

| Ref  | Proteus part | Value / setting | Every pin -> net          |
|------|--------------|-----------------|---------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND              |
| VIN1 | VSINE        | 50 mVpk, 1 kHz  | +=VSIG; -=GND             |
| CIN  | CAP          | 1 uF            | 1=VSIG; 2=GATE            |
| R1   | RES          | 1 MOhm          | 1=VDD; 2=GATE             |
| R2   | RES          | 220 kOhm        | 1=GATE; 2=GND             |
| RD   | RES          | 4.7 kOhm        | 1=VDD; 2=DRAIN            |
| RS   | RES          | 1 kOhm          | 1=SOURCE; 2=GND           |
| CS   | CAP          | 100 uF          | 1=SOURCE; 2=GND           |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=SOURCE |
| COUT | CAP          | 1 uF            | 1=DRAIN; 2=VOUT           |
| RL   | RES          | 10 kOhm         | 1=VOUT; 2=GND             |

## Net connection list

| Net    | Pins to join electrically              |
|--------|----------------------------------------|
| GND    | VDD1.-, VIN1.-, R2.2, RS.2, CS.2, RL.2 |
| VDD    | VDD1.+, R1.1, RD.1                     |
| DRAIN  | RD.2, Q1.D, COUT.1                     |
| GATE   | CIN.2, R1.2, R2.1, Q1.G                |
| SOURCE | RS.1, CS.1, Q1.S                       |
| VSIG   | VIN1.+, CIN.1                          |
| VOUT   | COUT.2, RL.1                           |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 101: NMOS Voltage-Controlled Resistor

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 4

| Ref   | Proteus part | Value / setting | Every pin -> net       |
|-------|--------------|-----------------|------------------------|
| VSIG  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND           |
| VCTRL | VDC          | 0-5 V sweep     | +=GATE; -=GND          |
| R1    | RES          | 1 kOhm          | 1=VIN; 2=DRAIN         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VSIG.-, VCTRL.-, Q1.S     |
| VIN   | VSIG.+, R1.1              |
| DRAIN | R1.2, Q1.D                |
| GATE  | VCTRL.+, Q1.G             |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 102: NMOS Constant-Current Sink

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 2 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net          |
|------|--------------|-----------------|---------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND              |
| VREF | VDC          | 2.5 V           | +=GATE; -=GND             |
| RL   | RES          | 100 Ohm         | 1=VDD; 2=DRAIN            |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=SOURCE |
| RS   | RES          | 10 Ohm          | 1=SOURCE; 2=GND           |

## Net connection list

| Net    | Pins to join electrically |
|--------|---------------------------|
| GND    | VDD1.-, VREF.-, RS.2      |
| VDD    | VDD1.+, RL.1              |
| DRAIN  | RL.2, Q1.D                |
| GATE   | VREF.+, Q1.G              |
| SOURCE | Q1.S, RS.1                |

PIN AUDIT: PASS - 11 of 11 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 103: NMOS Constant-Current LED Sink

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net          |
|------|--------------|-----------------|---------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND              |
| VREF | VDC          | 2.5 V           | +=GATE; -=GND             |
| LED1 | LED          | RED             | A=VDD; K=DRAIN            |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=SOURCE |
| RS   | RES          | 10 Ohm          | 1=SOURCE; 2=GND           |

## Net connection list

| Net    | Pins to join electrically |
|--------|---------------------------|
| GND    | VDD1.-, VREF.-, RS.2      |
| VDD    | VDD1.+, LED1.A            |
| DRAIN  | LED1.K, Q1.D              |
| GATE   | VREF.+, Q1.G              |
| SOURCE | Q1.S, RS.1                |

PIN AUDIT: PASS - 11 of 11 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 104: Two-NMOS Current Mirror

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 2 | Voltage source: 1 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net    |
|------|--------------|-----------------|---------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND        |
| RREF | RES          | 1 kOhm          | 1=VDD; 2=REF        |
| Q1   | NMOSFET      | Generic NMOS    | G=REF; D=REF; S=GND |
| Q2   | NMOSFET      | Generic NMOS    | G=REF; D=OUT; S=GND |
| RL   | RES          | 1 kOhm          | 1=VDD; 2=OUT        |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | VDD1.-, Q1.S, Q2.S        |
| VDD | VDD1.+, RREF.1, RL.1      |
| OUT | Q2.D, RL.2                |
| REF | RREF.2, Q1.G, Q1.D, Q2.G  |

PIN AUDIT: PASS - 12 of 12 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 105: NMOS Cascode Current Mirror

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 4 | Voltage source: 1 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net  |
|------|--------------|-----------------|-------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND      |
| RREF | RES          | 1 kOhm          | 1=VDD; 2=B2       |
| Q1   | NMOSFET      | Generic NMOS    | G=B1; D=B1; S=GND |
| Q2   | NMOSFET      | Generic NMOS    | G=B2; D=B2; S=B1  |
| Q3   | NMOSFET      | Generic NMOS    | G=B1; D=NX; S=GND |
| Q4   | NMOSFET      | Generic NMOS    | G=B2; D=OUT; S=NX |
| RL   | RES          | 1 kOhm          | 1=VDD; 2=OUT      |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | VDD1.-, Q1.S, Q3.S        |
| VDD | VDD1.+, RREF.1, RL.1      |
| B1  | Q1.G, Q1.D, Q2.S, Q3.G    |
| B2  | RREF.2, Q2.G, Q2.D, Q4.G  |
| NX  | Q3.D, Q4.S                |
| OUT | Q4.D, RL.2                |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 106: NMOS Amplifier with Active NMOS Load

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | NMOSFET: 2 | Voltage source: 2 | Total: 7

| Ref   | Proteus part | Value / setting | Every pin -> net         |
|-------|--------------|-----------------|--------------------------|
| VDD1  | VDC          | 12 V            | +=VDD; -=GND             |
| VIN1  | VSINE        | 50 mVpk, 1 kHz  | +=GATE; -=GND            |
| R1    | RES          | 1 MOhm          | 1=VDD; 2=GATE            |
| R2    | RES          | 220 kOhm        | 1=GATE; 2=GND            |
| QLOAD | NMOSFET      | Generic NMOS    | G=VDD; D=VDD; S=VOUT     |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=SOURCE |
| RS    | RES          | 1 kOhm          | 1=SOURCE; 2=GND          |

## Net connection list

| Net    | Pins to join electrically      |
|--------|--------------------------------|
| GND    | VDD1.-, VIN1.-, R2.2, RS.2     |
| VDD    | VDD1.+, R1.1, QLOAD.G, QLOAD.D |
| GATE   | VIN1.+, R1.2, R2.1, Q1.G       |
| SOURCE | Q1.S, RS.1                     |
| VOUT   | QLOAD.S, Q1.D                  |

PIN AUDIT: PASS - 16 of 16 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 107: NMOS Cascode Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 6 | NMOSFET: 2 | Voltage source: 2 | Total: 10

| Ref  | Proteus part | Value / setting | Every pin -> net      |
|------|--------------|-----------------|-----------------------|
| VDD1 | VDC          | 15 V            | +=VDD; -=GND          |
| VIN1 | VSINE        | 50 mVpk, 1 kHz  | +=GIN; -=GND          |
| R1   | RES          | 1 MOhm          | 1=VDD; 2=GIN          |
| R2   | RES          | 220 kOhm        | 1=GIN; 2=GND          |
| R3   | RES          | 470 kOhm        | 1=VDD; 2=GCAS         |
| R4   | RES          | 220 kOhm        | 1=GCAS; 2=GND         |
| RD   | RES          | 4.7 kOhm        | 1=VDD; 2=VOUT         |
| RS   | RES          | 1 kOhm          | 1=SOURCE; 2=GND       |
| Q1   | NMOSFET      | Generic NMOS    | G=GIN; D=N1; S=SOURCE |
| Q2   | NMOSFET      | Generic NMOS    | G=GCAS; D=VOUT; S=N1  |

## Net connection list

| Net    | Pins to join electrically        |
|--------|----------------------------------|
| GND    | VDD1.-, VIN1.-, R2.2, R4.2, RS.2 |
| VDD    | VDD1.+, R1.1, R3.1, RD.1         |
| GCAS   | R3.2, R4.1, Q2.G                 |
| GIN    | VIN1.+, R1.2, R2.1, Q1.G         |
| N1     | Q1.D, Q2.S                       |
| SOURCE | RS.1, Q1.S                       |
| VOUT   | RD.2, Q2.D                       |

PIN AUDIT: PASS - 22 of 22 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 108: Two-Stage NMOS Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 9 | Capacitor: 3 | NMOSFET: 2 | Voltage source: 2 | Total: 16

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| VDD1 | VDC          | 15 V            | +=VDD; -=GND     |
| VIN1 | VSINE        | 20 mVpk, 1 kHz  | +=VSIG; -=GND    |
| CIN  | CAP          | 1 uF            | 1=VSIG; 2=G1     |
| R1   | RES          | 1 MOhm          | 1=VDD; 2=G1      |
| R2   | RES          | 220 kOhm        | 1=G1; 2=GND      |
| RD1  | RES          | 4.7 kOhm        | 1=VDD; 2=D1      |
| RS1  | RES          | 1 kOhm          | 1=S1; 2=GND      |
| Q1   | NMOSFET      | Generic NMOS    | G=G1; D=D1; S=S1 |
| CC   | CAP          | 1 uF            | 1=D1; 2=G2       |
| R3   | RES          | 1 MOhm          | 1=VDD; 2=G2      |
| R4   | RES          | 220 kOhm        | 1=G2; 2=GND      |
| RD2  | RES          | 4.7 kOhm        | 1=VDD; 2=D2      |
| RS2  | RES          | 1 kOhm          | 1=S2; 2=GND      |
| Q2   | NMOSFET      | Generic NMOS    | G=G2; D=D2; S=S2 |
| COUT | CAP          | 1 uF            | 1=D2; 2=VOUT     |
| RL   | RES          | 10 kOhm         | 1=VOUT; 2=GND    |

## Net connection list

| Net  | Pins to join electrically                      |
|------|------------------------------------------------|
| GND  | VDD1.-, VIN1.-, R2.2, RS1.2, R4.2, RS2.2, RL.2 |
| VDD  | VDD1.+, R1.1, RD1.1, R3.1, RD2.1               |
| D1   | RD1.2, Q1.D, CC.1                              |
| D2   | RD2.2, Q2.D, COUT.1                            |
| G1   | CIN.2, R1.2, R2.1, Q1.G                        |
| G2   | CC.2, R3.2, R4.1, Q2.G                         |
| S1   | RS1.1, Q1.S                                    |
| S2   | RS2.1, Q2.S                                    |
| VSIG | VIN1.+, CIN.1                                  |
| VOUT | COUT.2, RL.1                                   |

PIN AUDIT: PASS - 34 of 34 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 109: Resistor-Load NMOS Inverter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 4

| Ref  | Proteus part | Value / setting | Every pin -> net      |
|------|--------------|-----------------|-----------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND          |
| VIN1 | VPULSE       | 0-5 V           | +=GATE; -=GND         |
| R1   | RES          | 10 kOhm         | 1=VDD; 2=VOUT         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=GND |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | VDD1.-, VIN1.-, Q1.S      |
| VDD  | VDD1.+, R1.1              |
| GATE | VIN1.+, Q1.G              |
| VOUT | R1.2, Q1.D                |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 110: Diode-Connected-Load NMOS Inverter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

NMOSFET: 2 | Voltage source: 2 | Total: 4

| Ref   | Proteus part | Value / setting | Every pin -> net      |
|-------|--------------|-----------------|-----------------------|
| VDD1  | VDC          | 5 V             | +=VDD; -=GND          |
| VIN1  | VPULSE       | 0-5 V           | +=GATE; -=GND         |
| QLOAD | NMOSFET      | Generic NMOS    | G=VDD; D=VDD; S=VOUT  |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=GND |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | VDD1.-, VIN1.-, Q1.S      |
| VDD  | VDD1.+, QLOAD.G, QLOAD.D  |
| GATE | VIN1.+, Q1.G              |
| VOUT | QLOAD.S, Q1.D             |

PIN AUDIT: PASS - 10 of 10 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 111: NMOS NAND Gate

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 2 | Voltage source: 3 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND     |
| VA   | VPULSE       | 0-5 V           | +=A; -=GND       |
| VB   | VPULSE       | 0-5 V           | +=B; -=GND       |
| R1   | RES          | 10 kOhm         | 1=VDD; 2=Y       |
| Q1   | NMOSFET      | Generic NMOS    | G=A; D=Y; S=N1   |
| Q2   | NMOSFET      | Generic NMOS    | G=B; D=N1; S=GND |

## Net connection list

| Net | Pins to join electrically |
|-----|---------------------------|
| GND | VDD1.-, VA.-, VB.-, Q2.S  |
| VDD | VDD1.+, R1.1              |
| A   | VA.+, Q1.G                |
| B   | VB.+, Q2.G                |
| N1  | Q1.S, Q2.D                |
| Y   | R1.2, Q1.D                |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 112: NMOS NOR Gate

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 2 | Voltage source: 3 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net |
|------|--------------|-----------------|------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND     |
| VA   | VPULSE       | 0-5 V           | +=A; -=GND       |
| VB   | VPULSE       | 0-5 V           | +=B; -=GND       |
| R1   | RES          | 10 kOhm         | 1=VDD; 2=Y       |
| Q1   | NMOSFET      | Generic NMOS    | G=A; D=Y; S=GND  |
| Q2   | NMOSFET      | Generic NMOS    | G=B; D=Y; S=GND  |

## Net connection list

| Net | Pins to join electrically      |
|-----|--------------------------------|
| GND | VDD1.-, VA.-, VB.-, Q1.S, Q2.S |
| VDD | VDD1.+, R1.1                   |
| A   | VA.+, Q1.G                     |
| B   | VB.+, Q2.G                     |
| Y   | R1.2, Q1.D, Q2.D               |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 113: NMOS Open-Drain Output

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 2 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net      |
|------|--------------|-----------------|-----------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND          |
| VIN1 | VPULSE       | 0-5 V           | +=GATE; -=GND         |
| RPU  | RES          | 10 kOhm         | 1=VDD; 2=VOUT         |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=GND |

## Net connection list

| Net  | Pins to join electrically   |
|------|-----------------------------|
| GND  | VDD1.-, VIN1.-, RPD.2, Q1.S |
| VDD  | VDD1.+, RPU.1               |
| GATE | VIN1.+, RPD.1, Q1.G         |
| VOUT | RPU.2, Q1.D                 |

PIN AUDIT: PASS - 11 of 11 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 114: NMOS Bidirectional Analog Switch

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 2 | Voltage source: 2 | Total: 6

| Ref   | Proteus part | Value / setting | Every pin -> net      |
|-------|--------------|-----------------|-----------------------|
| VSIG  | VSINE        | 1 Vpk, 1 kHz    | +=VIN; -=GND          |
| VCTRL | VDC          | 10 V            | +=GATE; -=GND         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=VIN; S=MID  |
| Q2    | NMOSFET      | Generic NMOS    | G=GATE; D=VOUT; S=MID |
| RPD   | RES          | 100 kOhm        | 1=GATE; 2=GND         |
| RL    | RES          | 10 kOhm         | 1=VOUT; 2=GND         |

## Net connection list

| Net  | Pins to join electrically    |
|------|------------------------------|
| GND  | VSIG.-, VCTRL.-, RPD.2, RL.2 |
| VIN  | VSIG.+, Q1.D                 |
| GATE | VCTRL.+, Q1.G, Q2.G, RPD.1   |
| MID  | Q1.S, Q2.S                   |
| VOUT | Q2.D, RL.1                   |

PIN AUDIT: PASS - 14 of 14 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 115: NMOS Level-Shifter Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 3 | Total: 6

| Ref   | Proteus part | Value / setting | Every pin -> net   |
|-------|--------------|-----------------|--------------------|
| VLOW  | VDC          | 3.3 V           | +=VDDL; -=GND      |
| VHIGH | VDC          | 12 V            | +=VDDH; -=GND      |
| VIN1  | VPULSE       | 0-3.3 V         | +=IN; -=GND        |
| R1    | RES          | 10 kOhm         | 1=VDDL; 2=IN       |
| R2    | RES          | 10 kOhm         | 1=VDDH; 2=OUT      |
| Q1    | NMOSFET      | Generic NMOS    | G=IN; D=OUT; S=GND |

## Net connection list

| Net  | Pins to join electrically     |
|------|-------------------------------|
| GND  | VLOW.-, VHIGH.-, VIN1.-, Q1.S |
| IN   | VIN1.+, R1.2, Q1.G            |
| OUT  | R2.2, Q1.D                    |
| VDDH | VHIGH.+, R2.1                 |
| VDDL | VLOW.+, R1.1                  |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 116: RC-Delayed NMOS Switch

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND           |
| VIN1 | VPULSE       | 0-5 V           | +=VIN; -=GND           |
| R1   | RES          | 100 kOhm        | 1=VIN; 2=GATE          |
| C1   | CAP          | 10 uF           | 1=GATE; 2=GND          |
| RL   | RES          | 1 kOhm          | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically  |
|-------|----------------------------|
| GND   | VDD1.-, VIN1.-, C1.2, Q1.S |
| VDD   | VDD1.+, RL.1               |
| VIN   | VIN1.+, R1.1               |
| DRAIN | RL.2, Q1.D                 |
| GATE  | R1.2, C1.1, Q1.G           |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 117: NMOS Soft-Start Load Switch

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref   | Proteus part | Value / setting | Every pin -> net       |
|-------|--------------|-----------------|------------------------|
| VDD1  | VDC          | 12 V            | +=VDD; -=GND           |
| VCTRL | VDC          | 5 V             | +=CTRL; -=GND          |
| R1    | RES          | 100 kOhm        | 1=CTRL; 2=GATE         |
| C1    | CAP          | 10 uF           | 1=GATE; 2=GND          |
| RL    | RES          | 1 kOhm          | 1=VDD; 2=DRAIN         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VDD1.-, VCTRL.-, C1.2, Q1.S |
| VDD   | VDD1.+, RL.1                |
| CTRL  | VCTRL.+, R1.1               |
| DRAIN | RL.2, Q1.D                  |
| GATE  | R1.2, C1.1, Q1.G            |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 118: NMOS LED Fade Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | LED: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND           |
| VIN1 | VPULSE       | 0-5 V, slow     | +=CTRL; -=GND          |
| R1   | RES          | 100 kOhm        | 1=CTRL; 2=GATE         |
| C1   | CAP          | 10 uF           | 1=GATE; 2=GND          |
| RLED | RES          | 330 Ohm         | 1=VDD; 2=LED_A         |
| LED1 | LED          | RED             | A=LED_A; K=DRAIN       |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically  |
|-------|----------------------------|
| GND   | VDD1.-, VIN1.-, C1.2, Q1.S |
| VDD   | VDD1.+, RLED.1             |
| CTRL  | VIN1.+, R1.1               |
| DRAIN | LED1.K, Q1.D               |
| GATE  | R1.2, C1.1, Q1.G           |
| LED_A | RLED.2, LED1.A             |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 119: NMOS PWM LED Dimmer

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | LED: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND           |
| VPWM | VPULSE       | 0-5 V PWM       | +=PWM; -=GND           |
| RG   | RES          | 100 Ohm         | 1=PWM; 2=GATE          |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| RLED | RES          | 330 Ohm         | 1=VDD; 2=LED_A         |
| LED1 | LED          | RED             | A=LED_A; K=DRAIN       |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VDD1.-, VPWM.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RLED.1              |
| DRAIN | LED1.K, Q1.D                |
| GATE  | RG.2, RPD.1, Q1.G           |
| LED_A | RLED.2, LED1.A              |
| PWM   | VPWM.+, RG.1                |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 120: NMOS Pulse Driver

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND           |
| VIN1 | VPULSE       | 0-5 V pulse     | +=DRV; -=GND           |
| RG   | RES          | 100 Ohm         | 1=DRV; 2=GATE          |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| RL   | RES          | 100 Ohm         | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VDD1.-, VIN1.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RL.1                |
| DRAIN | RL.2, Q1.D                  |
| DRV   | VIN1.+, RG.1                |
| GATE  | RG.2, RPD.1, Q1.G           |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 121: NMOS Inductor Pulse Driver

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 8

| Ref   | Proteus part | Value / setting | Every pin -> net       |
|-------|--------------|-----------------|------------------------|
| VDD1  | VDC          | 12 V            | +=VDD; -=GND           |
| VIN1  | VPULSE       | 0-5 V pulse     | +=DRV; -=GND           |
| RG    | RES          | 100 Ohm         | 1=DRV; 2=GATE          |
| RPD   | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| RCOIL | RES          | 100 Ohm         | 1=VDD; 2=N1            |
| L1    | INDUCTOR     | 100 mH          | 1=N1; 2=DRAIN          |
| D1    | DIODE        | 1N4007          | A=DRAIN; K=VDD         |
| Q1    | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VDD1.-, VIN1.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RCOIL.1, D1.K       |
| DRAIN | L1.2, D1.A, Q1.D            |
| DRV   | VIN1.+, RG.1                |
| GATE  | RG.2, RPD.1, Q1.G           |
| N1    | RCOIL.2, L1.1               |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 122: NMOS Capacitor-Discharge Switch

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VCHG | VDC          | 12 V            | +=VDD; -=GND           |
| VG   | VPULSE       | 0-5 V           | +=GATE; -=GND          |
| RCHG | RES          | 1 kOhm          | 1=VDD; 2=VCAP          |
| C1   | CAP          | 100 uF          | 1=VCAP; 2=GND          |
| RDIS | RES          | 10 Ohm          | 1=VCAP; 2=DRAIN        |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VCHG.-, VG.-, C1.2, Q1.S  |
| VDD   | VCHG.+, RCHG.1            |
| DRAIN | RDIS.2, Q1.D              |
| GATE  | VG.+, Q1.G                |
| VCAP  | RCHG.2, C1.1, RDIS.1      |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 123: NMOS Gate Pull-Down Discharge Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | NMOSFET: 1 | Voltage source: 2 | Total: 6

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND           |
| VIN1 | VPULSE       | 0-5 V           | +=DRV; -=GND           |
| RG   | RES          | 100 Ohm         | 1=DRV; 2=GATE          |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND          |
| RD   | RES          | 1 kOhm          | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VDD1.-, VIN1.-, RPD.2, Q1.S |
| VDD   | VDD1.+, RD.1                |
| DRAIN | RD.2, Q1.D                  |
| DRV   | VIN1.+, RG.1                |
| GATE  | RG.2, RPD.1, Q1.G           |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 124: NMOS Gate-Charge Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 2 | Total: 5

| Ref  | Proteus part | Value / setting   | Every pin -> net       |
|------|--------------|-------------------|------------------------|
| VDD1 | VDC          | 12 V              | +=VDD; -=GND           |
| VIN1 | VPULSE       | 0-10 V fast pulse | +=DRV; -=GND           |
| RG   | RES          | 10 Ohm            | 1=DRV; 2=GATE          |
| RD   | RES          | 10 Ohm            | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS      | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VIN1.-, Q1.S      |
| VDD   | VDD1.+, RD.1              |
| DRAIN | RD.2, Q1.D                |
| DRV   | VIN1.+, RG.1              |
| GATE  | RG.2, Q1.G                |

PIN AUDIT: PASS - 11 of 11 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 125: NMOS Transfer-Characteristic Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 4

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 10 V            | +=VDD; -=GND           |
| VGS  | VDC          | 0-5 V sweep     | +=GATE; -=GND          |
| RD   | RES          | 100 Ohm         | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VGS.-, Q1.S       |
| VDD   | VDD1.+, RD.1              |
| DRAIN | RD.2, Q1.D                |
| GATE  | VGS.+, Q1.G               |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 126: NMOS Output-Characteristic Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

NMOSFET: 1 | Voltage source: 2 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net       |
|-----|--------------|-----------------|------------------------|
| VDS | VDC          | 0-10 V sweep    | +=DRAIN; -=GND         |
| VGS | VDC          | 2-5 V stepped   | +=GATE; -=GND          |
| Q1  | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDS.-, VGS.-, Q1.S        |
| DRAIN | VDS.+, Q1.D               |
| GATE  | VGS.+, Q1.G               |

PIN AUDIT: PASS - 7 of 7 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 127: NMOS Threshold-Voltage Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 4

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 5 V             | +=VDD; -=GND           |
| VGS  | VDC          | 0-5 V sweep     | +=GATE; -=GND          |
| RD   | RES          | 100 kOhm        | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VGS.-, Q1.S       |
| VDD   | VDD1.+, RD.1              |
| DRAIN | RD.2, Q1.D                |
| GATE  | VGS.+, Q1.G               |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 128: NMOS On-Resistance Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

NMOSFET: 1 | Voltage source: 1 | Current source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net       |
|-----|--------------|-----------------|------------------------|
| VGS | VDC          | 10 V            | +=GATE; -=GND          |
| ID1 | IDC          | 1 A             | +=DRAIN; -=GND         |
| Q1  | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VGS.-, ID1.-, Q1.S        |
| DRAIN | ID1.+, Q1.D               |
| GATE  | VGS.+, Q1.G               |

PIN AUDIT: PASS - 7 of 7 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 129: NMOS Body-Diode Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | NMOSFET: 1 | Voltage source: 1 | Total: 3

| Ref | Proteus part | Value / setting | Every pin -> net            |
|-----|--------------|-----------------|-----------------------------|
| V1  | VDC          | 5 V             | +=SOURCE; -=GND             |
| R1  | RES          | 100 Ohm         | 1=DRAIN; 2=GND              |
| Q1  | NMOSFET      | Generic NMOS    | G=SOURCE; D=DRAIN; S=SOURCE |

## Net connection list

| Net    | Pins to join electrically |
|--------|---------------------------|
| GND    | V1.-, R1.2                |
| DRAIN  | R1.1, Q1.D                |
| SOURCE | V1.+, Q1.G, Q1.S          |

PIN AUDIT: PASS - 7 of 7 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 130: NMOS Switching-Loss Test Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | Voltage source: 2 | Total: 5

| Ref  | Proteus part | Value / setting | Every pin -> net       |
|------|--------------|-----------------|------------------------|
| VDD1 | VDC          | 12 V            | +=VDD; -=GND           |
| VIN1 | VPULSE       | 0-10 V, 100 kHz | +=DRV; -=GND           |
| RG   | RES          | 10 Ohm          | 1=DRV; 2=GATE          |
| RL   | RES          | 10 Ohm          | 1=VDD; 2=DRAIN         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=DRAIN; S=GND |

## Net connection list

| Net   | Pins to join electrically |
|-------|---------------------------|
| GND   | VDD1.-, VIN1.-, Q1.S      |
| VDD   | VDD1.+, RL.1              |
| DRAIN | RL.2, Q1.D                |
| DRV   | VIN1.+, RG.1              |
| GATE  | RG.2, Q1.G                |

PIN AUDIT: PASS - 11 of 11 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 131: Op-Amp Voltage Follower

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 5

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, 1 kHz        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=VOUT; OUT=VOUT; V+=V+; V-=V- |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically  |
|------|----------------------------|
| GND  | VCC.-, VEE.+, VIN1.-, RL.2 |
| V+   | VCC.+, U1.V+               |
| V-   | VEE.-, U1.V-               |
| VIN  | VIN1.+, U1.IN+             |
| VOUT | U1.IN-, U1.OUT, RL.1       |

PIN AUDIT: PASS - 13 of 13 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 132: Op-Amp Non-Inverting Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, 1 kHz        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RG   | RES          | 10 kOhm             | 1=NINV; 2=GND                             |
| RF   | RES          | 90 kOhm             | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically         |
|------|-----------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, RG.2, RL.2 |
| V+   | VCC.+ , U1.V+                     |
| V-   | VEE.-, U1.V-                      |
| VIN  | VIN1.+ , U1.IN+                   |
| NINV | U1.IN-, RG.1, RF.2                |
| VOUT | U1.OUT, RF.1, RL.1                |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 133: Op-Amp Inverting Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, 1 kHz        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RIN  | RES          | 10 kOhm             | 1=VIN; 2=NINV                             |
| RF   | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically           |
|------|-------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                       |
| V-   | VEE.-, U1.V-                        |
| VIN  | VIN1.+ , RIN.1                      |
| NINV | U1.IN-, RIN.2, RF.2                 |
| VOUT | U1.OUT, RF.1, RL.1                  |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 134: Potentiometer-Controlled Inverting Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | POT-HG: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, 1 kHz        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RIN  | RES          | 10 kOhm             | 1=VIN; 2=NINV                             |
| RV1  | POT-HG       | 100 kOhm            | 1=VOUT; W=NINV; 3=NINV                    |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically           |
|------|-------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                       |
| V-   | VEE.-, U1.V-                        |
| VIN  | VIN1.+ , RIN.1                      |
| NINV | U1.IN-, RIN.2, RV1.W, RV1.3         |
| VOUT | U1.OUT, RV1.1, RL.1                 |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 135: Potentiometer-Controlled Non-Inverting Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | POT-HG: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, 1 kHz        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RG   | RES          | 10 kOhm             | 1=NINV; 2=GND                             |
| RV1  | POT-HG       | 100 kOhm            | 1=VOUT; W=NINV; 3=NINV                    |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically         |
|------|-----------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, RG.2, RL.2 |
| V+   | VCC.+ , U1.V+                     |
| V-   | VEE.-, U1.V-                      |
| VIN  | VIN1.+ , U1.IN+                   |
| NINV | U1.IN-, RG.1, RV1.W, RV1.3        |
| VOUT | U1.OUT, RV1.1, RL.1               |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 136: Op-Amp Summing Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | OPAMP: 1 | Voltage source: 5 | Total: 11

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VDC          | 1 V                 | +=IN1; -=GND                             |
| VIN2 | VDC          | 2 V                 | +=IN2; -=GND                             |
| VIN3 | VDC          | 3 V                 | +=IN3; -=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=SUM; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 10 kOhm             | 1=IN1; 2=SUM                             |
| R2   | RES          | 10 kOhm             | 1=IN2; 2=SUM                             |
| R3   | RES          | 10 kOhm             | 1=IN3; 2=SUM                             |
| RF   | RES          | 10 kOhm             | 1=VOUT; 2=SUM                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                            |

## Net connection list

| Net  | Pins to join electrically                           |
|------|-----------------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, VIN2.-, VIN3.-, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                                       |
| V-   | VEE.-, U1.V-                                        |
| IN1  | VIN1.+ , R1.1                                       |
| IN2  | VIN2.+ , R2.1                                       |
| IN3  | VIN3.+ , R3.1                                       |
| SUM  | U1.IN-, R1.2, R2.2, R3.2, RF.2                      |
| VOUT | U1.OUT, RF.1, RL.1                                  |

PIN AUDIT: PASS - 25 of 25 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 137: Op-Amp Averaging Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | OPAMP: 1 | Voltage source: 5 | Total: 11

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VDC          | 1 V                 | +=IN1; -=GND                             |
| VIN2 | VDC          | 2 V                 | +=IN2; -=GND                             |
| VIN3 | VDC          | 3 V                 | +=IN3; -=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=SUM; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 30 kOhm             | 1=IN1; 2=SUM                             |
| R2   | RES          | 30 kOhm             | 1=IN2; 2=SUM                             |
| R3   | RES          | 30 kOhm             | 1=IN3; 2=SUM                             |
| RF   | RES          | 10 kOhm             | 1=VOUT; 2=SUM                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                            |

## Net connection list

| Net  | Pins to join electrically                           |
|------|-----------------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, VIN2.-, VIN3.-, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                                       |
| V-   | VEE.-, U1.V-                                        |
| IN1  | VIN1.+ , R1.1                                       |
| IN2  | VIN2.+ , R2.1                                       |
| IN3  | VIN3.+ , R3.1                                       |
| SUM  | U1.IN-, R1.2, R2.2, R3.2, RF.2                      |
| VOUT | U1.OUT, RF.1, RL.1                                  |

PIN AUDIT: PASS - 25 of 25 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 138: Op-Amp Difference Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | OPAMP: 1 | Voltage source: 4 | Total: 10

| Ref  | Proteus part | Value / setting     | Every pin -> net                              |
|------|--------------|---------------------|-----------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                   |
| VIN1 | VDC          | 1 V                 | +=IN1; -=GND                                  |
| VIN2 | VDC          | 2 V                 | +=IN2; -=GND                                  |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NMINUS; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 10 kOhm             | 1=IN1; 2=NMINUS                               |
| R2   | RES          | 10 kOhm             | 1=VOUT; 2=NMINUS                              |
| R3   | RES          | 10 kOhm             | 1=IN2; 2=NPLUS                                |
| R4   | RES          | 10 kOhm             | 1=NPLUS; 2=GND                                |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                                 |

## Net connection list

| Net    | Pins to join electrically                 |
|--------|-------------------------------------------|
| GND    | VCC.-, VEE.+ , VIN1.-, VIN2.-, R4.2, RL.2 |
| V+     | VCC.+ , U1.V+                             |
| V-     | VEE.-, U1.V-                              |
| IN1    | VIN1.+ , R1.1                             |
| IN2    | VIN2.+ , R3.1                             |
| NMINUS | U1.IN-, R1.2, R2.2                        |
| NPLUS  | U1.IN+ , R3.2, R4.1                       |
| VOUT   | U1.OUT, R2.1, RL.1                        |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 139: Three-Op-Amp Instrumentation Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 8 | OPAMP: 3 | Voltage source: 4 | Total: 15

| Ref  | Proteus part | Value / setting     | Every pin -> net                       |
|------|--------------|---------------------|----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                            |
| VEE  | VDC          | -15 V               | +=GND; -=V-                            |
| VIN1 | VDC          | 1 V                 | +=IN1; -=GND                           |
| VIN2 | VDC          | 1.1 V               | +=IN2; -=GND                           |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=IN1; IN-=N1; OUT=O1; V+=V+; V-=V-  |
| U2   | OPAMP        | Generic ideal OPAMP | IN+=IN2; IN-=N2; OUT=O2; V+=V+; V-=V-  |
| R1   | RES          | 10 kOhm             | 1=O1; 2=N1                             |
| R2   | RES          | 10 kOhm             | 1=O2; 2=N2                             |
| RG   | RES          | 1 kOhm              | 1=N1; 2=N2                             |
| U3   | OPAMP        | Generic ideal OPAMP | IN+=N4; IN-=N3; OUT=VOUT; V+=V+; V-=V- |
| R3   | RES          | 10 kOhm             | 1=O1; 2=N3                             |
| R4   | RES          | 10 kOhm             | 1=VOUT; 2=N3                           |
| R5   | RES          | 10 kOhm             | 1=O2; 2=N4                             |
| R6   | RES          | 10 kOhm             | 1=N4; 2=GND                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                          |

## Net connection list

| Net  | Pins to join electrically                 |
|------|-------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, VIN2.-, R6.2, RL.2 |
| V+   | VCC.+ , U1.V+ , U2.V+ , U3.V+             |
| V-   | VEE.-, U1.V- , U2.V- , U3.V-              |
| IN1  | VIN1.+ , U1.IN+                           |
| IN2  | VIN2.+ , U2.IN+                           |
| N1   | U1.IN- , R1.2, RG.1                       |
| N2   | U2.IN- , R2.2, RG.2                       |
| N3   | U3.IN- , R3.2, R4.2                       |
| N4   | U3.IN+ , R5.2, R6.1                       |
| O1   | U1.OUT , R1.1, R3.1                       |
| O2   | U2.OUT , R2.1, R5.1                       |
| VOUT | U3.OUT , R4.1, RL.1                       |

PIN AUDIT: PASS - 39 of 39 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 140: Differential-to-Single-Ended Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | OPAMP: 1 | Voltage source: 4 | Total: 10

| Ref  | Proteus part | Value / setting     | Every pin -> net                              |
|------|--------------|---------------------|-----------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                   |
| VINP | VDC          | 1.1 V               | +=INP; -=GND                                  |
| VINN | VDC          | 1.0 V               | +=INN; -=GND                                  |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NMINUS; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 10 kOhm             | 1=INN; 2=NMINUS                               |
| R2   | RES          | 10 kOhm             | 1=VOUT; 2=NMINUS                              |
| R3   | RES          | 10 kOhm             | 1=INP; 2=NPLUS                                |
| R4   | RES          | 10 kOhm             | 1=NPLUS; 2=GND                                |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                                 |

## Net connection list

| Net    | Pins to join electrically                 |
|--------|-------------------------------------------|
| GND    | VCC.-, VEE.+ , VINP.-, VINN.-, R4.2, RL.2 |
| V+     | VCC.+ , U1.V+                             |
| V-     | VEE.-, U1.V-                              |
| INN    | VINN.+ , R1.1                             |
| INP    | VINP.+ , R3.1                             |
| NMINUS | U1.IN-, R1.2, R2.2                        |
| NPLUS  | U1.IN+ , R3.2, R4.1                       |
| VOUT   | U1.OUT, R2.1, RL.1                        |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 141: AC-Coupled Inverting Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref   | Proteus part | Value / setting     | Every pin -> net                          |
|-------|--------------|---------------------|-------------------------------------------|
| VCC   | VDC          | +15 V               | +=V+; -=GND                               |
| VEE   | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1  | VSINE        | 1 Vpk, 1 kHz        | +=VSIG; -=GND                             |
| CIN   | CAP          | 1 uF                | 1=VSIG; 2=NAC                             |
| RBIAS | RES          | 100 kOhm            | 1=NAC; 2=GND                              |
| RIN   | RES          | 10 kOhm             | 1=NAC; 2=NINV                             |
| U1    | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RF    | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| RL    | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically                    |
|------|----------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, RBIAS.2, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                                |
| V-   | VEE.-, U1.V-                                 |
| NAC  | CIN.2, RBIAS.1, RIN.1                        |
| NINV | RIN.2, U1.IN-, RF.2                          |
| VSIG | VIN1.+ , CIN.1                               |
| VOUT | U1.OUT, RF.1, RL.1                           |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 142: AC-Coupled Non-Inverting Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref   | Proteus part | Value / setting     | Every pin -> net                            |
|-------|--------------|---------------------|---------------------------------------------|
| VCC   | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE   | VDC          | -15 V               | +=GND; -=V-                                 |
| VIN1  | VSINE        | 1 Vpk, 1 kHz        | +=VSIG; -=GND                               |
| CIN   | CAP          | 1 uF                | 1=VSIG; 2=NPLUS                             |
| RBIAS | RES          | 100 kOhm            | 1=NPLUS; 2=GND                              |
| U1    | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RG    | RES          | 10 kOhm             | 1=NINV; 2=GND                               |
| RF    | RES          | 90 kOhm             | 1=VOUT; 2=NINV                              |
| RL    | RES          | 10 kOhm             | 1=VOUT; 2=GND                               |

## Net connection list

| Net   | Pins to join electrically                  |
|-------|--------------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, RBIAS.2, RG.2, RL.2 |
| V+    | VCC.+ , U1.V+                              |
| V-    | VEE.-, U1.V-                               |
| NINV  | U1.IN-, RG.1, RF.2                         |
| NPLUS | CIN.2, RBIAS.1, U1.IN+                     |
| VSIG  | VIN1.+ , CIN.1                             |
| VOUT  | U1.OUT, RF.1, RL.1                         |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 143: Op-Amp Transimpedance Amplifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | OPAMP: 1 | Voltage source: 2 | Current source: 1 | Total: 7

| Ref | Proteus part | Value / setting     | Every pin -> net                          |
|-----|--------------|---------------------|-------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                               |
| VEE | VDC          | -15 V               | +=GND; -=V-                               |
| IIN | IDC          | 10 uA               | +=GND; -=NINV                             |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RF  | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| CF  | CAP          | 10 pF               | 1=VOUT; 2=NINV                            |
| RL  | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically            |
|------|--------------------------------------|
| GND  | VCC.-, VEE.+ , IIN.+ , U1.IN+ , RL.2 |
| V+   | VCC.+ , U1.V+                        |
| V-   | VEE.- , U1.V-                        |
| NINV | IIN.- , U1.IN- , RF.2 , CF.2         |
| VOUT | U1.OUT , RF.1 , CF.1 , RL.1          |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 144: Op-Amp Voltage-to-Current Converter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 6

| Ref  | Proteus part | Value / setting     | Every pin -> net                            |
|------|--------------|---------------------|---------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                 |
| VIN1 | VDC          | 1 V                 | +=VREF; -=GND                               |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VREF; IN-=SENSE; OUT=VOUT; V+=V+; V-=V- |
| RL   | RES          | 100 Ohm             | 1=VOUT; 2=SENSE                             |
| RS   | RES          | 10 Ohm              | 1=SENSE; 2=GND                              |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, RS.2 |
| V+    | VCC.+ , U1.V+               |
| V-    | VEE.-, U1.V-                |
| SENSE | U1.IN-, RL.2, RS.1          |
| VREF  | VIN1.+ , U1.IN+             |
| VOUT  | U1.OUT, RL.1                |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 145: Howland Current Pump

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                              |
|------|--------------|---------------------|-----------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                   |
| VIN1 | VDC          | 1 V                 | +=VIN; -=GND                                  |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NMINUS; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=NPLUS                                |
| R2   | RES          | 10 kOhm             | 1=NPLUS; 2=GND                                |
| R3   | RES          | 10 kOhm             | 1=VOUT; 2=NMINUS                              |
| R4   | RES          | 10 kOhm             | 1=NMINUS; 2=LOAD                              |
| RL   | RES          | 1 kOhm              | 1=LOAD; 2=GND                                 |

## Net connection list

| Net    | Pins to join electrically         |
|--------|-----------------------------------|
| GND    | VCC.-, VEE.+ , VIN1.-, R2.2, RL.2 |
| V+     | VCC.+ , U1.V+                     |
| V-     | VEE.-, U1.V-                      |
| VIN    | VIN1.+ , R1.1                     |
| LOAD   | R4.2, RL.1                        |
| NMINUS | U1.IN-, R3.2, R4.1                |
| NPLUS  | U1.IN+ , R1.2, R2.1               |
| VOUT   | U1.OUT, R3.1                      |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 146: Op-Amp Precision Current Source

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                           |
|------|--------------|---------------------|--------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                |
| VREF | VDC          | 1 V                 | +=REF; -=GND                               |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=REF; IN-=SENSE; OUT=GATE; V+=V+; V-=V- |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=V+; S=SOURCE                     |
| RS   | RES          | 10 Ohm              | 1=SOURCE; 2=SENSE                          |
| RL   | RES          | 100 Ohm             | 1=SENSE; 2=GND                             |

## Net connection list

| Net    | Pins to join electrically   |
|--------|-----------------------------|
| GND    | VCC.-, VEE.+ , VREF.-, RL.2 |
| V+     | VCC.+ , U1.V+ , Q1.D        |
| V-     | VEE.-, U1.V-                |
| GATE   | U1.OUT, Q1.G                |
| REF    | VREF.+ , U1.IN+             |
| SENSE  | U1.IN-, RS.2, RL.1          |
| SOURCE | Q1.S, RS.1                  |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 147: Op-Amp Precision Current Sink

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                           |
|------|--------------|---------------------|--------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                |
| VREF | VDC          | 1 V                 | +=REF; -=GND                               |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=REF; IN-=SENSE; OUT=GATE; V+=V+; V-=V- |
| RL   | RES          | 100 Ohm             | 1=V+; 2=DRAIN                              |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=DRAIN; S=SENSE                   |
| RS   | RES          | 10 Ohm              | 1=SENSE; 2=GND                             |

## Net connection list

| Net   | Pins to join electrically   |
|-------|-----------------------------|
| GND   | VCC.-, VEE.+ , VREF.-, RS.2 |
| V+    | VCC.+ , U1.V+ , RL.1        |
| V-    | VEE.-, U1.V-                |
| DRAIN | RL.2, Q1.D                  |
| GATE  | U1.OUT, Q1.G                |
| REF   | VREF.+ , U1.IN+             |
| SENSE | U1.IN-, Q1.S, RS.1          |

PIN AUDIT: PASS - 18 of 18 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 148: Op-Amp Comparator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 1 | OPAMP: 1 | Voltage source: 4 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VSINE        | 5 Vpk, 100 Hz       | +=VIN; -=GND                             |
| VREF | VDC          | 2 V                 | +=REF; -=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=REF; OUT=VOUT; V+=V+; V-=V- |
| RLED | RES          | 1 kOhm              | 1=VOUT; 2=LED_A                          |
| LED1 | LED          | RED                 | A=LED_A; K=GND                           |

## Net connection list

| Net   | Pins to join electrically             |
|-------|---------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, VREF.-, LED1.K |
| V+    | VCC.+ , U1.V+                         |
| V-    | VEE.-, U1.V-                          |
| VIN   | VIN1.+ , U1.IN+                       |
| LED_A | RLED.2, LED1.A                        |
| REF   | VREF.+ , U1.IN-                       |
| VOUT  | U1.OUT, RLED.1                        |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 149: Op-Amp Zero-Crossing Detector

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 1 | OPAMP: 1 | Voltage source: 3 | Total: 6

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VSINE        | 5 Vpk, 100 Hz       | +=VIN; -=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=GND; OUT=VOUT; V+=V+; V-=V- |
| RLED | RES          | 1 kOhm              | 1=VOUT; 2=LED_A                          |
| LED1 | LED          | RED                 | A=LED_A; K=GND                           |

## Net connection list

| Net   | Pins to join electrically             |
|-------|---------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, U1.IN-, LED1.K |
| V+    | VCC.+ , U1.V+                         |
| V-    | VEE.-, U1.V-                          |
| VIN   | VIN1.+ , U1.IN+                       |
| LED_A | RLED.2, LED1.A                        |
| VOUT  | U1.OUT, RLED.1                        |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 150: Op-Amp Level Detector

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 1 | OPAMP: 1 | Voltage source: 4 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VSINE        | 5 Vpk, 100 Hz       | +=VIN; -=GND                             |
| VREF | VDC          | 2.5 V               | +=REF; -=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=REF; OUT=VOUT; V+=V+; V-=V- |
| RLED | RES          | 1 kOhm              | 1=VOUT; 2=LED_A                          |
| LED1 | LED          | RED                 | A=LED_A; K=GND                           |

## Net connection list

| Net   | Pins to join electrically             |
|-------|---------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, VREF.-, LED1.K |
| V+    | VCC.+ , U1.V+                         |
| V-    | VEE.-, U1.V-                          |
| VIN   | VIN1.+ , U1.IN+                       |
| LED_A | RLED.2, LED1.A                        |
| REF   | VREF.+ , U1.IN-                       |
| VOUT  | U1.OUT, RLED.1                        |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 151: Dual-Op-Amp Window Comparator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Diode: 2 | LED: 1 | OPAMP: 2 | Voltage source: 5 | Total: 11

| Ref   | Proteus part | Value / setting     | Every pin -> net                        |
|-------|--------------|---------------------|-----------------------------------------|
| VCC   | VDC          | +15 V               | +=V+; -=GND                             |
| VEE   | VDC          | -15 V               | +=GND; -=V-                             |
| VIN1  | VSINE        | 5 Vpk, 100 Hz       | +=VIN; -=GND                            |
| VLOW  | VDC          | 1 V                 | +=LOW; -=GND                            |
| VHIGH | VDC          | 4 V                 | +=HIGH; -=GND                           |
| U1    | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=LOW; OUT=O1; V+=V+; V-=V-  |
| U2    | OPAMP        | Generic ideal OPAMP | IN+=HIGH; IN-=VIN; OUT=O2; V+=V+; V-=V- |
| RLED  | RES          | 1 kOhm              | 1=V+; 2=LED_A                           |
| LED1  | LED          | RED                 | A=LED_A; K=NODE                         |
| D1    | DIODE        | 1N4148              | A=NODE; K=O1                            |
| D2    | DIODE        | 1N4148              | A=NODE; K=O2                            |

## Net connection list

| Net   | Pins to join electrically              |
|-------|----------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, VLOW.-, VHIGH.- |
| V+    | VCC.+ , U1.V+ , U2.V+ , RLED.1         |
| V-    | VEE.-, U1.V- , U2.V-                   |
| VIN   | VIN1.+ , U1.IN+ , U2.IN-               |
| HIGH  | VHIGH.+ , U2.IN+                       |
| LED_A | RLED.2, LED1.A                         |
| LOW   | VLOW.+ , U1.IN-                        |
| NODE  | LED1.K, D1.A, D2.A                     |
| O1    | U1.OUT, D1.K                           |
| O2    | U2.OUT, D2.K                           |

PIN AUDIT: PASS - 28 of 28 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 152: Inverting Schmitt Trigger

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | LED: 1 | OPAMP: 1 | Voltage source: 3 | Total: 8

| Ref  | Proteus part | Value / setting     | Every pin -> net                           |
|------|--------------|---------------------|--------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                |
| VIN1 | VSINE        | 5 Vpk, 100 Hz       | +=VIN; -=GND                               |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=VIN; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 100 kOhm            | 1=VOUT; 2=NPLUS                            |
| R2   | RES          | 10 kOhm             | 1=NPLUS; 2=GND                             |
| RLED | RES          | 1 kOhm              | 1=VOUT; 2=LED_A                            |
| LED1 | LED          | RED                 | A=LED_A; K=GND                             |

## Net connection list

| Net   | Pins to join electrically           |
|-------|-------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, R2.2, LED1.K |
| V+    | VCC.+ , U1.V+                       |
| V-    | VEE.-, U1.V-                        |
| VIN   | VIN1.+ , U1.IN-                     |
| LED_A | RLED.2, LED1.A                      |
| NPLUS | U1.IN+ , R1.2, R2.1                 |
| VOUT  | U1.OUT, R1.1, RLED.1                |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 153: Non-Inverting Schmitt Trigger

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | LED: 1 | OPAMP: 1 | Voltage source: 3 | Total: 8

| Ref  | Proteus part | Value / setting     | Every pin -> net                           |
|------|--------------|---------------------|--------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                |
| VIN1 | VSINE        | 5 Vpk, 100 Hz       | +=VIN; -=GND                               |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=NPLUS                             |
| R2   | RES          | 100 kOhm            | 1=VOUT; 2=NPLUS                            |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=GND; OUT=VOUT; V+=V+; V-=V- |
| RLED | RES          | 1 kOhm              | 1=VOUT; 2=LED_A                            |
| LED1 | LED          | RED                 | A=LED_A; K=GND                             |

## Net connection list

| Net   | Pins to join electrically             |
|-------|---------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, U1.IN-, LED1.K |
| V+    | VCC.+ , U1.V+                         |
| V-    | VEE.-, U1.V-                          |
| VIN   | VIN1.+ , R1.1                         |
| LED_A | RLED.2, LED1.A                        |
| NPLUS | R1.2, R2.2, U1.IN+                    |
| VOUT  | R2.1, U1.OUT , RLED.1                 |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 154: Precision Half-Wave Rectifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Diode: 2 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                        |
|------|--------------|---------------------|-----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                             |
| VEE  | VDC          | -15 V               | +=GND; -=V-                             |
| VIN1 | VSINE        | 2 Vpk, 1 kHz        | +=VIN; -=GND                            |
| RIN  | RES          | 10 kOhm             | 1=VIN; 2=NINV                           |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=OA; V+=V+; V-=V- |
| D1   | DIODE        | 1N4148              | A=OA; K=VOUT                            |
| D2   | DIODE        | 1N4148              | A=NINV; K=OA                            |
| RF   | RES          | 10 kOhm             | 1=VOUT; 2=NINV                          |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                           |

## Net connection list

| Net  | Pins to join electrically            |
|------|--------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+ , RL.2 |
| V+   | VCC.+ , U1.V+                        |
| V-   | VEE.-, U1.V-                         |
| VIN  | VIN1.+ , RIN.1                       |
| NINV | RIN.2, U1.IN- , D2.A, RF.2           |
| OA   | U1.OUT, D1.A, D2.K                   |
| VOUT | D1.K, RF.1, RL.1                     |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 155: Precision Full-Wave Rectifier

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 6 | Diode: 4 | OPAMP: 2 | Voltage source: 3 | Total: 15

| Ref  | Proteus part | Value / setting     | Every pin -> net                                |
|------|--------------|---------------------|-------------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                     |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                     |
| VIN1 | VSINE        | 2 Vpk, 1 kHz        | +=VIN; -=GND                                    |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=N1; OUT=OA; V+=V+; V-=V-           |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=N1                                     |
| R2   | RES          | 10 kOhm             | 1=X; 2=N1                                       |
| D1   | DIODE        | 1N4148              | A=OA; K=P                                       |
| D2   | DIODE        | 1N4148              | A=N; K=OA                                       |
| D3   | DIODE        | 1N4148              | A=P; K=X                                        |
| D4   | DIODE        | 1N4148              | A=X; K=N                                        |
| R3   | RES          | 10 kOhm             | 1=P; 2=GND                                      |
| R4   | RES          | 10 kOhm             | 1=N; 2=GND                                      |
| U2   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS2; IN-=NMINUS2; OUT=VOUT; V+=V+; V-=V- |
| R5   | RES          | 10 kOhm             | 1=X; 2=NPLUS2                                   |
| R6   | RES          | 10 kOhm             | 1=VOUT; 2=NMINUS2                               |

## Net connection list

| Net     | Pins to join electrically                 |
|---------|-------------------------------------------|
| GND     | VCC.-, VEE.+ , VIN1.-, U1.IN+, R3.2, R4.2 |
| V+      | VCC.+ , U1.V+, U2.V+                      |
| V-      | VEE.-, U1.V-, U2.V-                       |
| VIN     | VIN1.+ , R1.1                             |
| N       | D2.A, D4.K, R4.1                          |
| N1      | U1.IN-, R1.2, R2.2                        |
| NMINUS2 | U2.IN-, R6.2                              |
| NPLUS2  | U2.IN+, R5.2                              |
| OA      | U1.OUT, D1.A, D2.K                        |
| P       | D1.K, D3.A, R3.1                          |
| X       | R2.1, D3.K, D4.A, R5.1                    |
| VOUT    | U2.OUT, R6.1                              |

PIN AUDIT: PASS - 36 of 36 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 156: Op-Amp Absolute-Value Circuit

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Diode: 2 | OPAMP: 2 | Voltage source: 3 | Total: 12

| Ref  | Proteus part | Value / setting     | Every pin -> net                                |
|------|--------------|---------------------|-------------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                     |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                     |
| VIN1 | VSINE        | 2 Vpk, 1 kHz        | +=VIN; -=GND                                    |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=N1; OUT=OA; V+=V+; V-=V-           |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=N1                                     |
| D1   | DIODE        | 1N4148              | A=OA; K=P                                       |
| D2   | DIODE        | 1N4148              | A=N1; K=OA                                      |
| R2   | RES          | 10 kOhm             | 1=P; 2=N1                                       |
| U2   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS2; IN-=NMINUS2; OUT=VOUT; V+=V+; V-=V- |
| R3   | RES          | 10 kOhm             | 1=VIN; 2=NPLUS2                                 |
| R4   | RES          | 10 kOhm             | 1=P; 2=NMINUS2                                  |
| R5   | RES          | 10 kOhm             | 1=VOUT; 2=NMINUS2                               |

## Net connection list

| Net     | Pins to join electrically     |
|---------|-------------------------------|
| GND     | VCC.-, VEE.+ , VIN1.-, U1.IN+ |
| V+      | VCC.+ , U1.V+ , U2.V+         |
| V-      | VEE.-, U1.V- , U2.V-          |
| VIN     | VIN1.+ , R1.1 , R3.1          |
| N1      | U1.IN- , R1.2 , D2.A , R2.2   |
| NMINUS2 | U2.IN- , R4.2 , R5.2          |
| NPLUS2  | U2.IN+ , R3.2                 |
| OA      | U1.OUT , D1.A , D2.K          |
| P       | D1.K , R2.1 , R4.1            |
| VOUT    | U2.OUT , R5.1                 |

PIN AUDIT: PASS - 30 of 30 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 157: Op-Amp Peak Detector

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | Capacitor: 1 | Diode: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                        |
|------|--------------|---------------------|-----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                             |
| VEE  | VDC          | -15 V               | +=GND; -=V-                             |
| VIN1 | VSINE        | 2 Vpk, 1 kHz        | +=VIN; -=GND                            |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=VOUT; OUT=OA; V+=V+; V-=V- |
| D1   | DIODE        | 1N4148              | A=OA; K=VOUT                            |
| C1   | CAP          | 1 uF                | 1=VOUT; 2=GND                           |
| RL   | RES          | 100 kOhm            | 1=VOUT; 2=GND                           |

## Net connection list

| Net  | Pins to join electrically         |
|------|-----------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, C1.2, RL.2 |
| V+   | VCC.+ , U1.V+                     |
| V-   | VEE.-, U1.V-                      |
| VIN  | VIN1.+ , U1.IN+                   |
| OA   | U1.OUT, D1.A                      |
| VOUT | U1.IN-, D1.K, C1.1, RL.1          |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 158: Op-Amp Active Clamp

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Diode: 1 | OPAMP: 1 | Voltage source: 4 | Total: 8

| Ref  | Proteus part | Value / setting     | Every pin -> net                        |
|------|--------------|---------------------|-----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                             |
| VEE  | VDC          | -15 V               | +=GND; -=V-                             |
| VIN1 | VSINE        | 5 Vpk, 1 kHz        | +=VIN; -=GND                            |
| VREF | VDC          | 2 V                 | +=REF; -=GND                            |
| R1   | RES          | 1 kOhm              | 1=VIN; 2=VOUT                           |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=REF; IN-=VOUT; OUT=OA; V+=V+; V-=V- |
| D1   | DIODE        | 1N4148              | A=OA; K=VOUT                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                           |

## Net connection list

| Net  | Pins to join electrically           |
|------|-------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, VREF.-, RL.2 |
| V+   | VCC.+ , U1.V+                       |
| V-   | VEE.-, U1.V-                        |
| VIN  | VIN1.+ , R1.1                       |
| OA   | U1.OUT, D1.A                        |
| REF  | VREF.+ , U1.IN+                     |
| VOUT | R1.2, U1.IN-, D1.K, RL.1            |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 159: Op-Amp Diode Limiter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Diode: 2 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, 1 kHz        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RIN  | RES          | 10 kOhm             | 1=VIN; 2=NINV                             |
| RF   | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| D1   | DIODE        | 1N4148              | A=VOUT; K=NINV                            |
| D2   | DIODE        | 1N4148              | A=NINV; K=VOUT                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically            |
|------|--------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+ , RL.2 |
| V+   | VCC.+ , U1.V+                        |
| V-   | VEE.-, U1.V-                         |
| VIN  | VIN1.+ , RIN.1                       |
| NINV | U1.IN-, RIN.2, RF.2, D1.K, D2.A      |
| VOUT | U1.OUT, RF.1, D1.A, D2.K, RL.1       |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 160: Ideal Op-Amp Integrator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VPULSE       | -1 to +1 V, 1 kHz   | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RIN  | RES          | 10 kOhm             | 1=VIN; 2=NINV                             |
| CF   | CAP          | 100 nF              | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically            |
|------|--------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+ , RL.2 |
| V+   | VCC.+ , U1.V+                        |
| V-   | VEE.-, U1.V-                         |
| VIN  | VIN1.+ , RIN.1                       |
| NINV | U1.IN-, RIN.2, CF.2                  |
| VOUT | U1.OUT, CF.1, RL.1                   |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 161: Practical Op-Amp Integrator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 8

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VPULSE       | -1 to +1 V, 1 kHz   | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RIN  | RES          | 10 kOhm             | 1=VIN; 2=NINV                             |
| CF   | CAP          | 100 nF              | 1=VOUT; 2=NINV                            |
| RF   | RES          | 1 MOhm              | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically            |
|------|--------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+ , RL.2 |
| V+   | VCC.+ , U1.V+                        |
| V-   | VEE.-, U1.V-                         |
| VIN  | VIN1.+ , RIN.1                       |
| NINV | U1.IN-, RIN.2, CF.2, RF.2            |
| VOUT | U1.OUT, CF.1, RF.1, RL.1             |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 162: Ideal Op-Amp Differentiator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 7

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VPULSE       | -1 to +1 V, 1 kHz   | +=VIN; -=GND                              |
| CIN  | CAP          | 10 nF               | 1=VIN; 2=NINV                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RF   | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically           |
|------|-------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                       |
| V-   | VEE.-, U1.V-                        |
| VIN  | VIN1.+ , CIN.1                      |
| NINV | CIN.2, U1.IN-, RF.2                 |
| VOUT | U1.OUT, RF.1, RL.1                  |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 163: Practical Op-Amp Differentiator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VPULSE       | -1 to +1 V, 1 kHz   | +=VIN; -=GND                              |
| CIN  | CAP          | 10 nF               | 1=VIN; 2=N1                               |
| RIN  | RES          | 1 kOhm              | 1=N1; 2=NINV                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RF   | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| CF   | CAP          | 100 pF              | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically           |
|------|-------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, RL.2 |
| V+   | VCC.+ , U1.V+                       |
| V-   | VEE.-, U1.V-                        |
| VIN  | VIN1.+ , CIN.1                      |
| N1   | CIN.2, RIN.1                        |
| NINV | RIN.2, U1.IN-, RF.2, CF.2           |
| VOUT | U1.OUT, RF.1, CF.1, RL.1            |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 164: First-Order Active Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                            |
|------|--------------|---------------------|---------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                 |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                                |
| R1   | RES          | 1.59 kOhm           | 1=VIN; 2=NPLUS                              |
| C1   | CAP          | 100 nF              | 1=NPLUS; 2=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RG   | RES          | 10 kOhm             | 1=NINV; 2=GND                               |
| RF   | RES          | 10 kOhm             | 1=VOUT; 2=NINV                              |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                               |

## Net connection list

| Net   | Pins to join electrically               |
|-------|-----------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, C1.2, RG.2, RL.2 |
| V+    | VCC.+ , U1.V+                           |
| V-    | VEE.-, U1.V-                            |
| VIN   | VIN1.+ , R1.1                           |
| NINV  | U1.IN-, RG.1, RF.2                      |
| NPLUS | R1.2, C1.1, U1.IN+                      |
| VOUT  | U1.OUT, RF.1, RL.1                      |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 165: First-Order Active High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                            |
|------|--------------|---------------------|---------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                 |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                                |
| C1   | CAP          | 100 nF              | 1=VIN; 2=NPLUS                              |
| R1   | RES          | 1.59 kOhm           | 1=NPLUS; 2=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RG   | RES          | 10 kOhm             | 1=NINV; 2=GND                               |
| RF   | RES          | 10 kOhm             | 1=VOUT; 2=NINV                              |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                               |

## Net connection list

| Net   | Pins to join electrically               |
|-------|-----------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, R1.2, RG.2, RL.2 |
| V+    | VCC.+ , U1.V+                           |
| V-    | VEE.-, U1.V-                            |
| VIN   | VIN1.+ , C1.1                           |
| NINV  | U1.IN-, RG.1, RF.2                      |
| NPLUS | C1.2, R1.1, U1.IN+                      |
| VOUT  | U1.OUT, RF.1, RL.1                      |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 166: Sallen-Key Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                             |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=N1                              |
| R2   | RES          | 10 kOhm             | 1=N1; 2=N2                               |
| C1   | CAP          | 10 nF               | 1=N1; 2=VOUT                             |
| C2   | CAP          | 10 nF               | 1=N2; 2=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=N2; IN-=VOUT; OUT=VOUT; V+=V+; V-=V- |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                            |

## Net connection list

| Net  | Pins to join electrically         |
|------|-----------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, C2.2, RL.2 |
| V+   | VCC.+ , U1.V+                     |
| V-   | VEE.-, U1.V-                      |
| VIN  | VIN1.+ , R1.1                     |
| N1   | R1.2, R2.1, C1.1                  |
| N2   | R2.2, C2.1, U1.IN+                |
| VOUT | C1.2, U1.IN-, U1.OUT, RL.1        |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 167: Sallen-Key High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                             |
| C1   | CAP          | 10 nF               | 1=VIN; 2=N1                              |
| C2   | CAP          | 10 nF               | 1=N1; 2=N2                               |
| R1   | RES          | 10 kOhm             | 1=N1; 2=VOUT                             |
| R2   | RES          | 10 kOhm             | 1=N2; 2=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=N2; IN-=VOUT; OUT=VOUT; V+=V+; V-=V- |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                            |

## Net connection list

| Net  | Pins to join electrically         |
|------|-----------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, R2.2, RL.2 |
| V+   | VCC.+ , U1.V+                     |
| V-   | VEE.-, U1.V-                      |
| VIN  | VIN1.+ , C1.1                     |
| N1   | C1.2, C2.1, R1.1                  |
| N2   | C2.2, R2.1, U1.IN+                |
| VOUT | R1.2, U1.IN-, U1.OUT, RL.1        |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 168: Multiple-Feedback Low-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 10

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=N1                               |
| R2   | RES          | 10 kOhm             | 1=N1; 2=NINV                              |
| RF   | RES          | 20 kOhm             | 1=VOUT; 2=NINV                            |
| C1   | CAP          | 10 nF               | 1=VOUT; 2=N1                              |
| C2   | CAP          | 10 nF               | 1=NINV; 2=GND                             |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically                 |
|------|-------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, C2.2, RL.2 |
| V+   | VCC.+ , U1.V+                             |
| V-   | VEE.-, U1.V-                              |
| VIN  | VIN1.+ , R1.1                             |
| N1   | R1.2, R2.1, C1.2                          |
| NINV | U1.IN-, R2.2, RF.2, C2.1                  |
| VOUT | U1.OUT, RF.1, C1.1, RL.1                  |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 169: Multiple-Feedback High-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 10

| Ref  | Proteus part | Value / setting     | Every pin -> net                          |
|------|--------------|---------------------|-------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                               |
| VEE  | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| C1   | CAP          | 10 nF               | 1=VIN; 2=N1                               |
| C2   | CAP          | 10 nF               | 1=N1; 2=NINV                              |
| R1   | RES          | 10 kOhm             | 1=N1; 2=GND                               |
| R2   | RES          | 10 kOhm             | 1=NINV; 2=GND                             |
| RF   | RES          | 20 kOhm             | 1=VOUT; 2=NINV                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically                       |
|------|-------------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, R1.2, R2.2, RL.2 |
| V+   | VCC.+ , U1.V+                                   |
| V-   | VEE.-, U1.V-                                    |
| VIN  | VIN1.+ , C1.1                                   |
| N1   | C1.2, C2.1, R1.1                                |
| NINV | U1.IN-, C2.2, R2.1, RF.2                        |
| VOUT | U1.OUT, RF.1, RL.1                              |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 170: Active Band-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 2 | OPAMP: 1 | Voltage source: 3 | Total: 10

| Ref   | Proteus part | Value / setting     | Every pin -> net                          |
|-------|--------------|---------------------|-------------------------------------------|
| VCC   | VDC          | +15 V               | +=V+; -=GND                               |
| VEE   | VDC          | -15 V               | +=GND; -=V-                               |
| VIN1  | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                              |
| C1    | CAP          | 10 nF               | 1=VIN; 2=N1                               |
| R1    | RES          | 10 kOhm             | 1=N1; 2=NINV                              |
| U1    | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| RF    | RES          | 100 kOhm            | 1=VOUT; 2=NINV                            |
| C2    | CAP          | 1 nF                | 1=VOUT; 2=NINV                            |
| RBIAS | RES          | 100 kOhm            | 1=N1; 2=GND                               |
| RL    | RES          | 10 kOhm             | 1=VOUT; 2=GND                             |

## Net connection list

| Net  | Pins to join electrically                    |
|------|----------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, U1.IN+, RBIAS.2, RL.2 |
| V+   | VCC.+ , U1.V+                                |
| V-   | VEE.-, U1.V-                                 |
| VIN  | VIN1.+ , C1.1                                |
| N1   | C1.2, R1.1, RBIAS.1                          |
| NINV | R1.2, U1.IN-, RF.2, C2.2                     |
| VOUT | U1.OUT, RF.1, C2.1, RL.1                     |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 171: Active Notch Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 6 | Capacitor: 3 | OPAMP: 1 | Voltage source: 3 | Total: 13

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                             |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=NRA                             |
| R2   | RES          | 10 kOhm             | 1=NRA; 2=NT                              |
| C3   | CAP          | 20 nF               | 1=NRA; 2=GND                             |
| C1   | CAP          | 10 nF               | 1=VIN; 2=NCA                             |
| C2   | CAP          | 10 nF               | 1=NCA; 2=NT                              |
| R3   | RES          | 5 kOhm              | 1=NCA; 2=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NT; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| R4   | RES          | 10 kOhm             | 1=VOUT; 2=NINV                           |
| R5   | RES          | 10 kOhm             | 1=NINV; 2=GND                            |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                            |

## Net connection list

| Net  | Pins to join electrically                     |
|------|-----------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, C3.2, R3.2, R5.2, RL.2 |
| V+   | VCC.+ , U1.V+                                 |
| V-   | VEE.-, U1.V-                                  |
| VIN  | VIN1.+ , R1.1, C1.1                           |
| NCA  | C1.2, C2.1, R3.1                              |
| NINV | U1.IN-, R4.2, R5.1                            |
| NRA  | R1.2, R2.1, C3.1                              |
| NT   | R2.2, C2.2, U1.IN+                            |
| VOUT | U1.OUT, R4.1, RL.1                            |

PIN AUDIT: PASS - 29 of 29 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 172: Active All-Pass Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 1 | OPAMP: 1 | Voltage source: 3 | Total: 9

| Ref  | Proteus part | Value / setting     | Every pin -> net                            |
|------|--------------|---------------------|---------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                 |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                                |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=NPLUS                              |
| C1   | CAP          | 10 nF               | 1=NPLUS; 2=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| R2   | RES          | 10 kOhm             | 1=VIN; 2=NINV                               |
| R3   | RES          | 10 kOhm             | 1=VOUT; 2=NINV                              |
| RL   | RES          | 10 kOhm             | 1=VOUT; 2=GND                               |

## Net connection list

| Net   | Pins to join electrically         |
|-------|-----------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, C1.2, RL.2 |
| V+    | VCC.+ , U1.V+                     |
| V-    | VEE.-, U1.V-                      |
| VIN   | VIN1.+ , R1.1, R2.1               |
| NINV  | U1.IN-, R2.2, R3.2                |
| NPLUS | R1.2, C1.1, U1.IN+                |
| VOUT  | U1.OUT, R3.1, RL.1                |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 173: State-Variable Filter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 8 | Capacitor: 2 | OPAMP: 3 | Voltage source: 3 | Total: 16

| Ref  | Proteus part | Value / setting     | Every pin -> net                       |
|------|--------------|---------------------|----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                            |
| VEE  | VDC          | -15 V               | +=GND; -=V-                            |
| VIN1 | VSINE        | 1 Vpk, sweep        | +=VIN; -=GND                           |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=SUM; OUT=HP; V+=V+; V-=V- |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=SUM                           |
| R2   | RES          | 10 kOhm             | 1=BP; 2=SUM                            |
| R3   | RES          | 10 kOhm             | 1=LP; 2=SUM                            |
| R4   | RES          | 10 kOhm             | 1=HP; 2=SUM                            |
| U2   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=I2; OUT=BP; V+=V+; V-=V-  |
| R5   | RES          | 10 kOhm             | 1=HP; 2=I2                             |
| C1   | CAP          | 10 nF               | 1=BP; 2=I2                             |
| U3   | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=I3; OUT=LP; V+=V+; V-=V-  |
| R6   | RES          | 10 kOhm             | 1=BP; 2=I3                             |
| C2   | CAP          | 10 nF               | 1=LP; 2=I3                             |
| R7   | RES          | 100 kOhm            | 1=BP; 2=GND                            |
| R8   | RES          | 100 kOhm            | 1=LP; 2=GND                            |

## Net connection list

| Net | Pins to join electrically                                 |
|-----|-----------------------------------------------------------|
| GND | VCC.-, VEE.+ , VIN1.-, U1.IN+, U2.IN+, U3.IN+, R7.2, R8.2 |
| V+  | VCC.+ , U1.V+, U2.V+, U3.V+                               |
| V-  | VEE.-, U1.V-, U2.V-, U3.V-                                |
| VIN | VIN1.+ , R1.1                                             |
| BP  | R2.1, U2.OUT, C1.1, R6.1, R7.1                            |
| HP  | U1.OUT, R4.1, R5.1                                        |
| I2  | U2.IN-, R5.2, C1.2                                        |
| I3  | U3.IN-, R6.2, C2.2                                        |
| LP  | R3.1, U3.OUT, C2.1, R8.1                                  |
| SUM | U1.IN-, R1.2, R2.2, R3.2, R4.2                            |

PIN AUDIT: PASS - 41 of 41 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 174: Wien-Bridge Oscillator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Capacitor: 2 | Diode: 2 | OPAMP: 1 | Voltage source: 2 | Total: 11

| Ref | Proteus part | Value / setting     | Every pin -> net                            |
|-----|--------------|---------------------|---------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE | VDC          | -15 V               | +=GND; -=V-                                 |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| C1  | CAP          | 10 nF               | 1=VOUT; 2=NSER                              |
| R1  | RES          | 10 kOhm             | 1=NSER; 2=NPLUS                             |
| R2  | RES          | 10 kOhm             | 1=NPLUS; 2=GND                              |
| C2  | CAP          | 10 nF               | 1=NPLUS; 2=GND                              |
| R3  | RES          | 20 kOhm             | 1=VOUT; 2=NINV                              |
| R4  | RES          | 10 kOhm             | 1=NINV; 2=GND                               |
| D1  | DIODE        | 1N4148              | A=VOUT; K=NINV                              |
| D2  | DIODE        | 1N4148              | A=NINV; K=VOUT                              |

## Net connection list

| Net   | Pins to join electrically       |
|-------|---------------------------------|
| GND   | VCC.-, VEE.+ , R2.2, C2.2, R4.2 |
| V+    | VCC.+ , U1.V+                   |
| V-    | VEE.-, U1.V-                    |
| NINV  | U1.IN-, R3.2, R4.1, D1.K, D2.A  |
| NPLUS | U1.IN+, R1.2, R2.1, C2.1        |
| NSER  | C1.2, R1.1                      |
| VOUT  | U1.OUT, C1.1, R3.1, D1.A, D2.K  |

PIN AUDIT: PASS - 25 of 25 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 175: Op-Amp Phase-Shift Oscillator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Capacitor: 3 | OPAMP: 1 | Voltage source: 2 | Total: 11

| Ref | Proteus part | Value / setting     | Every pin -> net                          |
|-----|--------------|---------------------|-------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                               |
| VEE | VDC          | -15 V               | +=GND; -=V-                               |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV; OUT=VOUT; V+=V+; V-=V- |
| C1  | CAP          | 10 nF               | 1=VOUT; 2=N1                              |
| R1  | RES          | 10 kOhm             | 1=N1; 2=GND                               |
| C2  | CAP          | 10 nF               | 1=N1; 2=N2                                |
| R2  | RES          | 10 kOhm             | 1=N2; 2=GND                               |
| C3  | CAP          | 10 nF               | 1=N2; 2=N3                                |
| R3  | RES          | 10 kOhm             | 1=N3; 2=GND                               |
| R4  | RES          | 10 kOhm             | 1=N3; 2=NINV                              |
| R5  | RES          | 300 kOhm            | 1=VOUT; 2=NINV                            |

## Net connection list

| Net  | Pins to join electrically                |
|------|------------------------------------------|
| GND  | VCC.-, VEE.+ , U1.IN+ , R1.2, R2.2, R3.2 |
| V+   | VCC.+ , U1.V+                            |
| V-   | VEE.- , U1.V-                            |
| N1   | C1.2, R1.1, C2.1                         |
| N2   | C2.2, R2.1, C3.1                         |
| N3   | C3.2, R3.1, R4.1                         |
| NINV | U1.IN- , R4.2, R5.2                      |
| VOUT | U1.OUT, C1.1, R5.1                       |

PIN AUDIT: PASS - 25 of 25 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 176: Op-Amp Relaxation Oscillator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | OPAMP: 1 | Voltage source: 2 | Total: 7

| Ref | Proteus part | Value / setting     | Every pin -> net                              |
|-----|--------------|---------------------|-----------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE | VDC          | -15 V               | +=GND; -=V-                                   |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NMINUS; OUT=VOUT; V+=V+; V-=V- |
| R1  | RES          | 100 kOhm            | 1=VOUT; 2=NPLUS                               |
| R2  | RES          | 10 kOhm             | 1=NPLUS; 2=GND                                |
| R3  | RES          | 100 kOhm            | 1=VOUT; 2=NMINUS                              |
| C1  | CAP          | 10 nF               | 1=NMINUS; 2=GND                               |

## Net connection list

| Net    | Pins to join electrically |
|--------|---------------------------|
| GND    | VCC.-, VEE.+ , R2.2, C1.2 |
| V+     | VCC.+ , U1.V+             |
| V-     | VEE.-, U1.V-              |
| NMINUS | U1.IN-, R3.2, C1.1        |
| NPLUS  | U1.IN+, R1.2, R2.1        |
| VOUT   | U1.OUT, R1.1, R3.1        |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 177: Op-Amp Square-Wave Generator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | OPAMP: 1 | Voltage source: 2 | Total: 7

| Ref | Proteus part | Value / setting     | Every pin -> net                              |
|-----|--------------|---------------------|-----------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE | VDC          | -15 V               | +=GND; -=V-                                   |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=NPLUS; IN-=NMINUS; OUT=VOUT; V+=V+; V-=V- |
| R1  | RES          | 100 kOhm            | 1=VOUT; 2=NPLUS                               |
| R2  | RES          | 10 kOhm             | 1=NPLUS; 2=GND                                |
| R3  | RES          | 100 kOhm            | 1=VOUT; 2=NMINUS                              |
| C1  | CAP          | 10 nF               | 1=NMINUS; 2=GND                               |

## Net connection list

| Net    | Pins to join electrically |
|--------|---------------------------|
| GND    | VCC.-, VEE.+ , R2.2, C1.2 |
| V+     | VCC.+ , U1.V+             |
| V-     | VEE.- , U1.V-             |
| NMINUS | U1.IN-, R3.2, C1.1        |
| NPLUS  | U1.IN+ , R1.2, R2.1       |
| VOUT   | U1.OUT, R1.1, R3.1        |

PIN AUDIT: PASS - 17 of 17 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 178: Op-Amp Triangle-Wave Generator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Capacitor: 1 | OPAMP: 2 | Voltage source: 2 | Total: 10

| Ref | Proteus part | Value / setting     | Every pin -> net                              |
|-----|--------------|---------------------|-----------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE | VDC          | -15 V               | +=GND; -=V-                                   |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=NPLUS1; IN-=TRI; OUT=SQUARE; V+=V+; V-=V- |
| R1  | RES          | 100 kOhm            | 1=SQUARE; 2=NPLUS1                            |
| R2  | RES          | 10 kOhm             | 1=NPLUS1; 2=GND                               |
| U2  | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV2; OUT=TRI; V+=V+; V-=V-     |
| R3  | RES          | 100 kOhm            | 1=SQUARE; 2=NINV2                             |
| C1  | CAP          | 10 nF               | 1=TRI; 2=NINV2                                |
| R4  | RES          | 10 kOhm             | 1=SQUARE; 2=GND                               |
| R5  | RES          | 10 kOhm             | 1=TRI; 2=GND                                  |

## Net connection list

| Net    | Pins to join electrically               |
|--------|-----------------------------------------|
| GND    | VCC.-, VEE.+ , R2.2, U2.IN+, R4.2, R5.2 |
| V+     | VCC.+ , U1.V+ , U2.V+                   |
| V-     | VEE.-, U1.V-, U2.V-                     |
| NINV2  | U2.IN-, R3.2, C1.2                      |
| NPLUS1 | U1.IN+, R1.2, R2.1                      |
| SQUARE | U1.OUT, R1.1, R3.1, R4.1                |
| TRI    | U1.IN-, U2.OUT, C1.1, R5.1              |

PIN AUDIT: PASS - 26 of 26 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 179: Op-Amp Sawtooth-Wave Generator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Capacitor: 1 | Diode: 1 | OPAMP: 2 | Voltage source: 2 | Total: 11

| Ref | Proteus part | Value / setting     | Every pin -> net                              |
|-----|--------------|---------------------|-----------------------------------------------|
| VCC | VDC          | +15 V               | +=V+; -=GND                                   |
| VEE | VDC          | -15 V               | +=GND; -=V-                                   |
| U1  | OPAMP        | Generic ideal OPAMP | IN+=NPLUS1; IN-=SAW; OUT=SQUARE; V+=V+; V-=V- |
| R1  | RES          | 100 kOhm            | 1=SQUARE; 2=NPLUS1                            |
| R2  | RES          | 10 kOhm             | 1=NPLUS1; 2=GND                               |
| U2  | OPAMP        | Generic ideal OPAMP | IN+=GND; IN-=NINV2; OUT=SAW; V+=V+; V-=V-     |
| R3  | RES          | 100 kOhm            | 1=SQUARE; 2=NINV2                             |
| D1  | DIODE        | 1N4148              | A=SQUARE; K=NINV2                             |
| C1  | CAP          | 10 nF               | 1=SAW; 2=NINV2                                |
| R4  | RES          | 10 kOhm             | 1=SQUARE; 2=GND                               |
| R5  | RES          | 10 kOhm             | 1=SAW; 2=GND                                  |

## Net connection list

| Net    | Pins to join electrically                |
|--------|------------------------------------------|
| GND    | VCC.-, VEE.+ , R2.2, U2.IN+ , R4.2, R5.2 |
| V+     | VCC.+ , U1.V+ , U2.V+                    |
| V-     | VEE.- , U1.V- , U2.V-                    |
| NINV2  | U2.IN- , R3.2, D1.K, C1.2                |
| NPLUS1 | U1.IN+ , R1.2, R2.1                      |
| SAW    | U1.IN- , U2.OUT , C1.1, R5.1             |
| SQUARE | U1.OUT , R1.1, R3.1, D1.A, R4.1          |

PIN AUDIT: PASS - 28 of 28 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 180: Op-Amp LED Level Indicator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 8 | LED: 4 | OPAMP: 4 | Voltage source: 3 | Total: 19

| Ref   | Proteus part | Value / setting     | Every pin -> net                       |
|-------|--------------|---------------------|----------------------------------------|
| VCC   | VDC          | +15 V               | +=V+; -=GND                            |
| VEE   | VDC          | -15 V               | +=GND; -=V-                            |
| VIN1  | VDC          | 0-15 V sweep        | +=VIN; -=GND                           |
| R1    | RES          | 10 kOhm             | 1=V+; 2=T1                             |
| R2    | RES          | 10 kOhm             | 1=T1; 2=T2                             |
| R3    | RES          | 10 kOhm             | 1=T2; 2=T3                             |
| R4    | RES          | 10 kOhm             | 1=T3; 2=GND                            |
| U1    | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=GND; OUT=O1; V+=V+; V-=V- |
| RLED1 | RES          | 1 kOhm              | 1=O1; 2=A1                             |
| LED1  | LED          | RED                 | A=A1; K=GND                            |
| U2    | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=T3; OUT=O2; V+=V+; V-=V-  |
| RLED2 | RES          | 1 kOhm              | 1=O2; 2=A2                             |
| LED2  | LED          | YELLOW              | A=A2; K=GND                            |
| U3    | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=T2; OUT=O3; V+=V+; V-=V-  |
| RLED3 | RES          | 1 kOhm              | 1=O3; 2=A3                             |
| LED3  | LED          | GREEN               | A=A3; K=GND                            |
| U4    | OPAMP        | Generic ideal OPAMP | IN+=VIN; IN-=T1; OUT=O4; V+=V+; V-=V-  |
| RLED4 | RES          | 1 kOhm              | 1=O4; 2=A4                             |
| LED4  | LED          | BLUE                | A=A4; K=GND                            |

## Net connection list

| Net | Pins to join electrically                                           |
|-----|---------------------------------------------------------------------|
| GND | VCC.-, VEE.+ , VIN1.-, R4.2, U1.IN-, LED1.K, LED2.K, LED3.K, LED4.K |
| V+  | VCC.+ , R1.1, U1.V+, U2.V+, U3.V+, U4.V+                            |
| V-  | VEE.-, U1.V-, U2.V-, U3.V-, U4.V-                                   |
| VIN | VIN1.+ , U1.IN+, U2.IN+, U3.IN+, U4.IN+                             |
| A1  | RLED1.2, LED1.A                                                     |
| A2  | RLED2.2, LED2.A                                                     |
| A3  | RLED3.2, LED3.A                                                     |
| A4  | RLED4.2, LED4.A                                                     |
| O1  | U1.OUT, RLED1.1                                                     |
| O2  | U2.OUT, RLED2.1                                                     |
| O3  | U3.OUT, RLED3.1                                                     |
| O4  | U4.OUT, RLED4.1                                                     |
| T1  | R1.2, R2.1, U4.IN-                                                  |
| T2  | R2.2, R3.1, U3.IN-                                                  |
| T3  | R3.2, R4.1, U2.IN-                                                  |

PIN AUDIT: PASS - 50 of 50 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 181: LM317 Adjustable Voltage Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | Capacitor: 1 | Electrolytic capacitor: 1 | POT-HG: 1 | LM317T: 1 | Voltage source: 1 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net                   |
|------|--------------|-----------------|------------------------------------|
| V1   | VDC          | 15 V            | +=VIN; -=GND                       |
| U1   | LM317T       | LM317T          | ADJ(1)=ADJ; OUT(2)=VOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm         | 1=VOUT; 2=ADJ                      |
| RV1  | POT-HG       | 5 kOhm          | 1=ADJ; W=GND; 3=GND                |
| CIN  | CAP          | 100 nF          | 1=VIN; 2=GND                       |
| COUT | CAP-ELEC     | 10 uF           | +=VOUT; -=GND                      |
| RL   | RES          | 1 kOhm          | 1=VOUT; 2=GND                      |

## Net connection list

| Net  | Pins to join electrically               |
|------|-----------------------------------------|
| GND  | V1.-, RV1.W, RV1.3, CIN.2, COUT.-, RL.2 |
| VIN  | V1.+, U1.IN(3), CIN.1                   |
| ADJ  | U1.ADJ(1), R1.2, RV1.1                  |
| VOUT | U1.OUT(2), R1.1, COUT.+, RL.1           |

PIN AUDIT: PASS - 16 of 16 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 182: LM317 Fixed 5 V Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 1 | LM317T: 1 | Voltage source: 1 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net                   |
|------|--------------|-----------------|------------------------------------|
| V1   | VDC          | 12 V            | +=VIN; -=GND                       |
| U1   | LM317T       | LM317T          | ADJ(1)=ADJ; OUT(2)=VOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm         | 1=VOUT; 2=ADJ                      |
| R2   | RES          | 720 Ohm         | 1=ADJ; 2=GND                       |
| CIN  | CAP          | 100 nF          | 1=VIN; 2=GND                       |
| COUT | CAP-ELEC     | 10 uF           | +=VOUT; -=GND                      |
| RL   | RES          | 1 kOhm          | 1=VOUT; 2=GND                      |

## Net connection list

| Net  | Pins to join electrically       |
|------|---------------------------------|
| GND  | V1.-, R2.2, CIN.2, COUT.-, RL.2 |
| VIN  | V1.+, U1.IN(3), CIN.1           |
| ADJ  | U1.ADJ(1), R1.2, R2.1           |
| VOUT | U1.OUT(2), R1.1, COUT.+, RL.1   |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 183: LM317 Fixed 12 V Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 1 | LM317T: 1 | Voltage source: 1 | Total: 7

| Ref  | Proteus part | Value / setting | Every pin -> net                   |
|------|--------------|-----------------|------------------------------------|
| V1   | VDC          | 18 V            | +=VIN; -=GND                       |
| U1   | LM317T       | LM317T          | ADJ(1)=ADJ; OUT(2)=VOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm         | 1=VOUT; 2=ADJ                      |
| R2   | RES          | 2.06 kOhm       | 1=ADJ; 2=GND                       |
| CIN  | CAP          | 100 nF          | 1=VIN; 2=GND                       |
| COUT | CAP-ELEC     | 10 uF           | +=VOUT; -=GND                      |
| RL   | RES          | 1 kOhm          | 1=VOUT; 2=GND                      |

## Net connection list

| Net  | Pins to join electrically       |
|------|---------------------------------|
| GND  | V1.-, R2.2, CIN.2, COUT.-, RL.2 |
| VIN  | V1.+, U1.IN(3), CIN.1           |
| ADJ  | U1.ADJ(1), R1.2, R2.1           |
| VOUT | U1.OUT(2), R1.1, COUT.+, RL.1   |

PIN AUDIT: PASS - 15 of 15 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 184: LM317 Constant-Current Source

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | LM317T: 1 | Voltage source: 1 | Total: 4

| Ref  | Proteus part | Value / setting | Every pin -> net                    |
|------|--------------|-----------------|-------------------------------------|
| V1   | VDC          | 15 V            | +=VIN; -=GND                        |
| U1   | LM317T       | LM317T          | ADJ(1)=LOAD; OUT(2)=NOUT; IN(3)=VIN |
| RSET | RES          | 12.5 Ohm        | 1=NOUT; 2=LOAD                      |
| RL   | RES          | 100 Ohm         | 1=LOAD; 2=GND                       |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, RL.2                |
| VIN  | V1.+, U1.IN(3)            |
| LOAD | U1.ADJ(1), RSET.2, RL.1   |
| NOUT | U1.OUT(2), RSET.1         |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 185: LM317 Constant-Current LED Driver

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 1 | LED: 1 | LM317T: 1 | Voltage source: 1 | Total: 4

| Ref  | Proteus part | Value / setting | Every pin -> net                    |
|------|--------------|-----------------|-------------------------------------|
| V1   | VDC          | 12 V            | +=VIN; -=GND                        |
| U1   | LM317T       | LM317T          | ADJ(1)=LOAD; OUT(2)=NOUT; IN(3)=VIN |
| RSET | RES          | 62 Ohm          | 1=NOUT; 2=LOAD                      |
| LED1 | LED          | WHITE           | A=LOAD; K=GND                       |

## Net connection list

| Net  | Pins to join electrically |
|------|---------------------------|
| GND  | V1.-, LED1.K              |
| VIN  | V1.+, U1.IN(3)            |
| LOAD | U1.ADJ(1), RSET.2, LED1.A |
| NOUT | U1.OUT(2), RSET.1         |

PIN AUDIT: PASS - 9 of 9 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 186: LM317 Trickle Battery Charger

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 1 | Diode: 1 | LM317T: 1 | Voltage source: 2 | Total: 9

| Ref  | Proteus part | Value / setting    | Every pin -> net                     |
|------|--------------|--------------------|--------------------------------------|
| VSUP | VDC          | 15 V               | +=VIN; -=GND                         |
| VBAT | VDC          | 12 V battery model | +=BAT; -=GND                         |
| U1   | LM317T       | LM317T             | ADJ(1)=ADJ; OUT(2)=REGOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm            | 1=REGOUT; 2=ADJ                      |
| R2   | RES          | 2.06 kOhm          | 1=ADJ; 2=GND                         |
| RCHG | RES          | 100 Ohm            | 1=REGOUT; 2=BAT                      |
| D1   | DIODE        | 1N4007             | A=BAT; K=REGOUT                      |
| CIN  | CAP          | 100 nF             | 1=VIN; 2=GND                         |
| COUT | CAP-ELEC     | 10 uF              | +=REGOUT; -=GND                      |

## Net connection list

| Net    | Pins to join electrically             |
|--------|---------------------------------------|
| GND    | VSUP.-, VBAT.-, R2.2, CIN.2, COUT.-   |
| VIN    | VSUP.+, U1.IN(3), CIN.1               |
| ADJ    | U1.ADJ(1), R1.2, R2.1                 |
| BAT    | VBAT.+, RCHG.2, D1.A                  |
| REGOUT | U1.OUT(2), R1.1, RCHG.1, D1.K, COUT.+ |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 187: LM317 Current-Limited Battery Charger

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 1 | POT-HG: 1 | Diode: 1 | LM317T: 1 | Voltage source: 2 | Total: 10

| Ref    | Proteus part | Value / setting    | Every pin -> net                     |
|--------|--------------|--------------------|--------------------------------------|
| VSUP   | VDC          | 15 V               | +=VIN; -=GND                         |
| VBAT   | VDC          | 12 V battery model | +=BAT; -=GND                         |
| U1     | LM317T       | LM317T             | ADJ(1)=ADJ; OUT(2)=REGOUT; IN(3)=VIN |
| R1     | RES          | 240 Ohm            | 1=REGOUT; 2=ADJ                      |
| RV1    | POT-HG       | 5 kOhm             | 1=ADJ; W=GND; 3=GND                  |
| RCHG   | RES          | 47 Ohm             | 1=REGOUT; 2=BAT                      |
| RBLEED | RES          | 10 kOhm            | 1=BAT; 2=GND                         |
| D1     | DIODE        | 1N4007             | A=BAT; K=REGOUT                      |
| CIN    | CAP          | 100 nF             | 1=VIN; 2=GND                         |
| COUT   | CAP-ELEC     | 10 uF              | +=REGOUT; -=GND                      |

## Net connection list

| Net    | Pins to join electrically                             |
|--------|-------------------------------------------------------|
| GND    | VSUP.-, VBAT.-, RV1.W, RV1.3, RBLEED.2, CIN.2, COUT.- |
| VIN    | VSUP.+, U1.IN(3), CIN.1                               |
| ADJ    | U1.ADJ(1), R1.2, RV1.1                                |
| BAT    | VBAT.+, RCHG.2, RBLEED.1, D1.A                        |
| REGOUT | U1.OUT(2), R1.1, RCHG.1, D1.K, COUT.+                 |

PIN AUDIT: PASS - 22 of 22 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 188: LM317 Ripple-Rejection-Enhanced Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 2 | Diode: 2 | LM317T: 1 | Voltage source: 1 | Total: 10

| Ref  | Proteus part | Value / setting | Every pin -> net                   |
|------|--------------|-----------------|------------------------------------|
| V1   | VDC          | 15 V            | +=VIN; -=GND                       |
| U1   | LM317T       | LM317T          | ADJ(1)=ADJ; OUT(2)=VOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm         | 1=VOUT; 2=ADJ                      |
| R2   | RES          | 2 kOhm          | 1=ADJ; 2=GND                       |
| RL   | RES          | 1 kOhm          | 1=VOUT; 2=GND                      |
| CIN  | CAP          | 100 nF          | 1=VIN; 2=GND                       |
| COUT | CAP-ELEC     | 10 uF           | +=VOUT; -=GND                      |
| CADJ | CAP-ELEC     | 10 uF           | +=ADJ; -=GND                       |
| D1   | DIODE        | 1N4007          | A=VOUT; K=VIN                      |
| D2   | DIODE        | 1N4007          | A=ADJ; K=VOUT                      |

## Net connection list

| Net  | Pins to join electrically                 |
|------|-------------------------------------------|
| GND  | V1.-, R2.2, RL.2, CIN.2, COUT.-, CADJ.-   |
| VIN  | V1.+, U1.IN(3), CIN.1, D1.K               |
| ADJ  | U1.ADJ(1), R1.2, R2.1, CADJ.+, D2.A       |
| VOUT | U1.OUT(2), R1.1, RL.1, COUT.+, D1.A, D2.K |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 189: LM317 Slow-Start Voltage Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 2 | Diode: 1 | LM317T: 1 | Voltage source: 1 | Total: 9

| Ref  | Proteus part | Value / setting | Every pin -> net                   |
|------|--------------|-----------------|------------------------------------|
| V1   | VDC          | 15 V            | +=VIN; -=GND                       |
| U1   | LM317T       | LM317T          | ADJ(1)=ADJ; OUT(2)=VOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm         | 1=VOUT; 2=ADJ                      |
| R2   | RES          | 2 kOhm          | 1=ADJ; 2=GND                       |
| RL   | RES          | 1 kOhm          | 1=VOUT; 2=GND                      |
| CIN  | CAP          | 100 nF          | 1=VIN; 2=GND                       |
| COUT | CAP-ELEC     | 10 uF           | +=VOUT; -=GND                      |
| CSS  | CAP-ELEC     | 100 uF          | +=ADJ; -=GND                       |
| D1   | DIODE        | 1N4007          | A=ADJ; K=VOUT                      |

## Net connection list

| Net  | Pins to join electrically              |
|------|----------------------------------------|
| GND  | V1.-, R2.2, RL.2, CIN.2, COUT.-, CSS.- |
| VIN  | V1.+, U1.IN(3), CIN.1                  |
| ADJ  | U1.ADJ(1), R1.2, R2.1, CSS.+, D1.A     |
| VOUT | U1.OUT(2), R1.1, RL.1, COUT.+, D1.K    |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 190: LM317 Reverse-Discharge-Protected Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 1 | Diode: 2 | LM317T: 1 | Voltage source: 1 | Total: 9

| Ref  | Proteus part | Value / setting | Every pin -> net                   |
|------|--------------|-----------------|------------------------------------|
| V1   | VDC          | 15 V            | +=VIN; -=GND                       |
| U1   | LM317T       | LM317T          | ADJ(1)=ADJ; OUT(2)=VOUT; IN(3)=VIN |
| R1   | RES          | 240 Ohm         | 1=VOUT; 2=ADJ                      |
| R2   | RES          | 2 kOhm          | 1=ADJ; 2=GND                       |
| RL   | RES          | 1 kOhm          | 1=VOUT; 2=GND                      |
| CIN  | CAP          | 100 nF          | 1=VIN; 2=GND                       |
| COUT | CAP-ELEC     | 100 uF          | +=VOUT; -=GND                      |
| D1   | DIODE        | 1N4007          | A=VOUT; K=VIN                      |
| D2   | DIODE        | 1N4007          | A=ADJ; K=VOUT                      |

## Net connection list

| Net  | Pins to join electrically                 |
|------|-------------------------------------------|
| GND  | V1.-, R2.2, RL.2, CIN.2, COUT.-           |
| VIN  | V1.+, U1.IN(3), CIN.1, D1.K               |
| ADJ  | U1.ADJ(1), R1.2, R2.1, D2.A               |
| VOUT | U1.OUT(2), R1.1, RL.1, COUT.+, D1.A, D2.K |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 191: Op-Amp-Controlled NMOS Electronic Load

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 2 | POT-HG: 1 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 8

| Ref   | Proteus part | Value / setting     | Every pin -> net                         |
|-------|--------------|---------------------|------------------------------------------|
| VCC   | VDC          | +15 V               | +=V+; -=GND                              |
| VEE   | VDC          | -15 V               | +=GND; -=V-                              |
| VLOAD | VDC          | 12 V                | +=LOADP; -=GND                           |
| RV1   | POT-HG       | 10 kOhm             | 1=V+; W=SET; 3=GND                       |
| U1    | OPAMP        | Generic ideal OPAMP | IN+=SET; IN-=SENSE; OUT=OG; V+=V+; V-=V- |
| RG    | RES          | 100 Ohm             | 1=OG; 2=GATE                             |
| Q1    | NMOSFET      | Generic NMOS        | G=GATE; D=LOADP; S=SENSE                 |
| RS    | RES          | 1 Ohm               | 1=SENSE; 2=GND                           |

## Net connection list

| Net   | Pins to join electrically           |
|-------|-------------------------------------|
| GND   | VCC.-, VEE.+ , VLOAD.-, RV1.3, RS.2 |
| V+    | VCC.+ , RV1.1, U1.V+                |
| V-    | VEE.-, U1.V-                        |
| GATE  | RG.2, Q1.G                          |
| LOADP | VLOAD.+ , Q1.D                      |
| OG    | U1.OUT, RG.1                        |
| SENSE | U1.IN-, Q1.S, RS.1                  |
| SET   | RV1.W, U1.IN+                       |

PIN AUDIT: PASS - 21 of 21 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 192: Op-Amp-Controlled NMOS Linear Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 4 | Electrolytic capacitor: 1 | POT-HG: 1 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 11

| Ref  | Proteus part | Value / setting     | Every pin -> net                      |
|------|--------------|---------------------|---------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                           |
| VEE  | VDC          | -15 V               | +=GND; -=V-                           |
| VIN1 | VDC          | 20 V                | +=VIN; -=GND                          |
| RV1  | POT-HG       | 10 kOhm             | 1=V+; W=SET; 3=GND                    |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=SET; IN-=FB; OUT=OG; V+=V+; V-=V- |
| RG   | RES          | 100 Ohm             | 1=OG; 2=GATE                          |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=VIN; S=VOUT                 |
| R1   | RES          | 10 kOhm             | 1=VOUT; 2=FB                          |
| R2   | RES          | 10 kOhm             | 1=FB; 2=GND                           |
| COUT | CAP-ELEC     | 100 uF              | +=VOUT; -=GND                         |
| RL   | RES          | 1 kOhm              | 1=VOUT; 2=GND                         |

## Net connection list

| Net  | Pins to join electrically                        |
|------|--------------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, RV1.3, R2.2, COUT.-, RL.2 |
| V+   | VCC.+ , RV1.1, U1.V+                             |
| V-   | VEE.-, U1.V-                                     |
| VIN  | VIN1.+ , Q1.D                                    |
| FB   | U1.IN-, R1.2, R2.1                               |
| GATE | RG.2, Q1.G                                       |
| OG   | U1.OUT, RG.1                                     |
| SET  | RV1.W, U1.IN+                                    |
| VOUT | Q1.S, R1.1, COUT.+ , RL.1                        |

PIN AUDIT: PASS - 27 of 27 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 193: Zener-Referenced Op-Amp NMOS Regulator

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Electrolytic capacitor: 1 | Zener: 1 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 12

| Ref  | Proteus part | Value / setting     | Every pin -> net                       |
|------|--------------|---------------------|----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                            |
| VEE  | VDC          | -15 V               | +=GND; -=V-                            |
| VIN1 | VDC          | 20 V                | +=VIN; -=GND                           |
| RZ   | RES          | 1 kOhm              | 1=VIN; 2=VREF                          |
| DZ1  | ZENER        | 5.1 V               | A=GND; K=VREF                          |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VREF; IN-=FB; OUT=OG; V+=V+; V-=V- |
| RG   | RES          | 100 Ohm             | 1=OG; 2=GATE                           |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=VIN; S=VOUT                  |
| R1   | RES          | 10 kOhm             | 1=VOUT; 2=FB                           |
| R2   | RES          | 10 kOhm             | 1=FB; 2=GND                            |
| COUT | CAP-ELEC     | 100 uF              | +=VOUT; -=GND                          |
| RL   | RES          | 1 kOhm              | 1=VOUT; 2=GND                          |

## Net connection list

| Net  | Pins to join electrically                        |
|------|--------------------------------------------------|
| GND  | VCC.-, VEE.+ , VIN1.-, DZ1.A, R2.2, COUT.-, RL.2 |
| V+   | VCC.+ , U1.V+                                    |
| V-   | VEE.-, U1.V-                                     |
| VIN  | VIN1.+ , RZ.1, Q1.D                              |
| FB   | U1.IN-, R1.2, R2.1                               |
| GATE | RG.2, Q1.G                                       |
| OG   | U1.OUT, RG.1                                     |
| VREF | RZ.2, DZ1.K, U1.IN+                              |
| VOUT | Q1.S, R1.1, COUT.+ , RL.1                        |

PIN AUDIT: PASS - 28 of 28 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 194: Zener Comparator NMOS Overvoltage Cutoff

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | Zener: 1 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 11

| Ref  | Proteus part | Value / setting     | Every pin -> net                            |
|------|--------------|---------------------|---------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                                 |
| VEE  | VDC          | -15 V               | +=GND; -=V-                                 |
| VIN1 | VDC          | 0-20 V sweep        | +=VIN; -=GND                                |
| RZ   | RES          | 1 kOhm              | 1=VIN; 2=VREF                               |
| DZ1  | ZENER        | 5.1 V               | A=GND; K=VREF                               |
| R1   | RES          | 10 kOhm             | 1=VIN; 2=SENSE                              |
| R2   | RES          | 3.3 kOhm            | 1=SENSE; 2=GND                              |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=VREF; IN-=SENSE; OUT=GATE; V+=V+; V-=V- |
| RH   | RES          | 100 kOhm            | 1=GATE; 2=SENSE                             |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=VIN; S=VOUT                       |
| RL   | RES          | 1 kOhm              | 1=VOUT; 2=GND                               |

## Net connection list

| Net   | Pins to join electrically                |
|-------|------------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, DZ1.A, R2.2, RL.2 |
| V+    | VCC.+ , U1.V+                            |
| V-    | VEE.-, U1.V-                             |
| VIN   | VIN1.+ , RZ.1, R1.1, Q1.D                |
| GATE  | U1.OUT, RH.1, Q1.G                       |
| SENSE | R1.2, R2.1, U1.IN-, RH.2                 |
| VREF  | RZ.2, DZ1.K, U1.IN+                      |
| VOUT  | Q1.S, RL.1                               |

PIN AUDIT: PASS - 26 of 26 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 195: Op-Amp NMOS Undervoltage Lockout

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 5 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 4 | Total: 11

| Ref  | Proteus part | Value / setting     | Every pin -> net                         |
|------|--------------|---------------------|------------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                              |
| VEE  | VDC          | -15 V               | +=GND; -=V-                              |
| VIN1 | VDC          | 0-20 V sweep        | +=VIN; -=GND                             |
| VREF | VDC          | 2.5 V               | +=REF; -=GND                             |
| R1   | RES          | 100 kOhm            | 1=VIN; 2=SENSE                           |
| R2   | RES          | 22 kOhm             | 1=SENSE; 2=GND                           |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=SENSE; IN-=REF; OUT=OG; V+=V+; V-=V- |
| RH   | RES          | 1 MOhm              | 1=OG; 2=SENSE                            |
| RG   | RES          | 100 Ohm             | 1=OG; 2=GATE                             |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=VIN; S=VOUT                    |
| RL   | RES          | 1 kOhm              | 1=VOUT; 2=GND                            |

## Net connection list

| Net   | Pins to join electrically                 |
|-------|-------------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, VREF.-, R2.2, RL.2 |
| V+    | VCC.+ , U1.V+                             |
| V-    | VEE.-, U1.V-                              |
| VIN   | VIN1.+ , R1.1, Q1.D                       |
| GATE  | RG.2, Q1.G                                |
| OG    | U1.OUT, RH.1, RG.1                        |
| REF   | VREF.+ , U1.IN-                           |
| SENSE | R1.2, R2.1, U1.IN+ , RH.2                 |
| VOUT  | Q1.S, RL.1                                |

PIN AUDIT: PASS - 26 of 26 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 196: Op-Amp NMOS Electronic Fuse

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 6 | Zener: 1 | NMOSFET: 1 | OPAMP: 1 | Voltage source: 3 | Total: 12

| Ref  | Proteus part | Value / setting     | Every pin -> net                        |
|------|--------------|---------------------|-----------------------------------------|
| VCC  | VDC          | +15 V               | +=V+; -=GND                             |
| VEE  | VDC          | -15 V               | +=GND; -=V-                             |
| VIN1 | VDC          | 12 V                | +=VIN; -=GND                            |
| RZ   | RES          | 1 kOhm              | 1=VIN; 2=VREF                           |
| DZ1  | ZENER        | 2.7 V               | A=GND; K=VREF                           |
| RL   | RES          | 10 Ohm              | 1=VIN; 2=DRAIN                          |
| Q1   | NMOSFET      | Generic NMOS        | G=GATE; D=DRAIN; S=SENSE                |
| RS   | RES          | 0.1 Ohm             | 1=SENSE; 2=GND                          |
| R1   | RES          | 100 kOhm            | 1=VREF; 2=TH                            |
| R2   | RES          | 10 kOhm             | 1=TH; 2=GND                             |
| U1   | OPAMP        | Generic ideal OPAMP | IN+=TH; IN-=SENSE; OUT=OG; V+=V+; V-=V- |
| RG   | RES          | 100 Ohm             | 1=OG; 2=GATE                            |

## Net connection list

| Net   | Pins to join electrically                |
|-------|------------------------------------------|
| GND   | VCC.-, VEE.+ , VIN1.-, DZ1.A, RS.2, R2.2 |
| V+    | VCC.+ , U1.V+                            |
| V-    | VEE.-, U1.V-                             |
| VIN   | VIN1.+ , RZ.1, RL.1                      |
| DRAIN | RL.2, Q1.D                               |
| GATE  | Q1.G, RG.2                               |
| OG    | U1.OUT, RG.1                             |
| SENSE | Q1.S, RS.1, U1.IN-                       |
| TH    | R1.2, R2.1, U1.IN+                       |
| VREF  | RZ.2, DZ1.K, R1.1                        |

PIN AUDIT: PASS - 28 of 28 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 197: Open-Loop NMOS Buck Converter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Electrolytic capacitor: 1 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 9

| Ref  | Proteus part | Value / setting | Every pin -> net    |
|------|--------------|-----------------|---------------------|
| VIN1 | VDC          | 12 V            | +=VIN; -=GND        |
| VPWM | VPULSE       | 0-15 V PWM      | +=DRV; -=GND        |
| RG   | RES          | 10 Ohm          | 1=DRV; 2=GATE       |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND       |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=VIN; S=SW |
| D1   | DIODE        | 1N5819          | A=GND; K=SW         |
| L1   | INDUCTOR     | 100 uH          | 1=SW; 2=VOUT        |
| COUT | CAP-ELEC     | 220 uF          | +=VOUT; -=GND       |
| RL   | RES          | 10 Ohm          | 1=VOUT; 2=GND       |

## Net connection list

| Net  | Pins to join electrically                 |
|------|-------------------------------------------|
| GND  | VIN1.-, VPWM.-, RPD.2, D1.A, COUT.-, RL.2 |
| VIN  | VIN1.+, Q1.D                              |
| DRV  | VPWM.+, RG.1                              |
| GATE | RG.2, RPD.1, Q1.G                         |
| SW   | Q1.S, D1.K, L1.1                          |
| VOUT | L1.2, COUT.+, RL.1                        |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 198: Open-Loop NMOS Boost Converter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Electrolytic capacitor: 1 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 9

| Ref  | Proteus part | Value / setting | Every pin -> net    |
|------|--------------|-----------------|---------------------|
| VIN1 | VDC          | 5 V             | +=VIN; -=GND        |
| VPWM | VPULSE       | 0-10 V PWM      | +=DRV; -=GND        |
| RG   | RES          | 10 Ohm          | 1=DRV; 2=GATE       |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND       |
| L1   | INDUCTOR     | 100 uH          | 1=VIN; 2=SW         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=SW; S=GND |
| D1   | DIODE        | 1N5819          | A=SW; K=VOUT        |
| COUT | CAP-ELEC     | 220 uF          | +=VOUT; -=GND       |
| RL   | RES          | 100 Ohm         | 1=VOUT; 2=GND       |

## Net connection list

| Net  | Pins to join electrically                 |
|------|-------------------------------------------|
| GND  | VIN1.-, VPWM.-, RPD.2, Q1.S, COUT.-, RL.2 |
| VIN  | VIN1.+, L1.1                              |
| DRV  | VPWM.+, RG.1                              |
| GATE | RG.2, RPD.1, Q1.G                         |
| SW   | L1.2, Q1.D, D1.A                          |
| VOUT | D1.K, COUT.+, RL.1                        |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 199: Open-Loop NMOS Inverting Buck-Boost Converter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Electrolytic capacitor: 1 | Inductor: 1 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 9

| Ref  | Proteus part | Value / setting | Every pin -> net    |
|------|--------------|-----------------|---------------------|
| VIN1 | VDC          | 5 V             | +=VIN; -=GND        |
| VPWM | VPULSE       | 0-10 V PWM      | +=DRV; -=GND        |
| RG   | RES          | 10 Ohm          | 1=DRV; 2=GATE       |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND       |
| L1   | INDUCTOR     | 100 uH          | 1=VIN; 2=SW         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=SW; S=GND |
| D1   | DIODE        | 1N5819          | A=VOUTN; K=SW       |
| COUT | CAP-ELEC     | 220 uF          | +=GND; -=VOUTN      |
| RL   | RES          | 100 Ohm         | 1=GND; 2=VOUTN      |

## Net connection list

| Net   | Pins to join electrically                 |
|-------|-------------------------------------------|
| GND   | VIN1.-, VPWM.-, RPD.2, Q1.S, COUT.+, RL.1 |
| VIN   | VIN1.+, L1.1                              |
| DRV   | VPWM.+, RG.1                              |
| GATE  | RG.2, RPD.1, Q1.G                         |
| SW    | L1.2, Q1.D, D1.K                          |
| VOUTN | D1.A, COUT.-, RL.2                        |

PIN AUDIT: PASS - 19 of 19 placed component pins assigned to nets; 0 unassigned pins.

# Circuit 200: Open-Loop NMOS SEPIC Converter

Proteus rule: pins shown on the same named net are connected together. GND is a Proteus ground terminal and is not included in the component count.

Resistor: 3 | Capacitor: 1 | Electrolytic capacitor: 1 | Inductor: 2 | Diode: 1 | NMOSFET: 1 | Voltage source: 2 | Total: 11

| Ref  | Proteus part | Value / setting | Every pin -> net    |
|------|--------------|-----------------|---------------------|
| VIN1 | VDC          | 5 V             | +=VIN; -=GND        |
| VPWM | VPULSE       | 0-10 V PWM      | +=DRV; -=GND        |
| RG   | RES          | 10 Ohm          | 1=DRV; 2=GATE       |
| RPD  | RES          | 100 kOhm        | 1=GATE; 2=GND       |
| L1   | INDUCTOR     | 100 uH          | 1=VIN; 2=SW         |
| Q1   | NMOSFET      | Generic NMOS    | G=GATE; D=SW; S=GND |
| C1   | CAP          | 10 uF           | 1=SW; 2=N2          |
| L2   | INDUCTOR     | 100 uH          | 1=N2; 2=GND         |
| D1   | DIODE        | 1N5819          | A=N2; K=VOUT        |
| COUT | CAP-ELEC     | 220 uF          | +=VOUT; -=GND       |
| RL   | RES          | 100 Ohm         | 1=VOUT; 2=GND       |

## Net connection list

| Net  | Pins to join electrically                       |
|------|-------------------------------------------------|
| GND  | VIN1.-, VPWM.-, RPD.2, Q1.S, L2.2, COUT.-, RL.2 |
| VIN  | VIN1.+, L1.1                                    |
| DRV  | VPWM.+, RG.1                                    |
| GATE | RG.2, RPD.1, Q1.G                               |
| N2   | C1.2, L2.1, D1.A                                |
| SW   | L1.2, Q1.D, C1.1                                |
| VOUT | D1.K, COUT.+, RL.1                              |

PIN AUDIT: PASS - 23 of 23 placed component pins assigned to nets; 0 unassigned pins.